

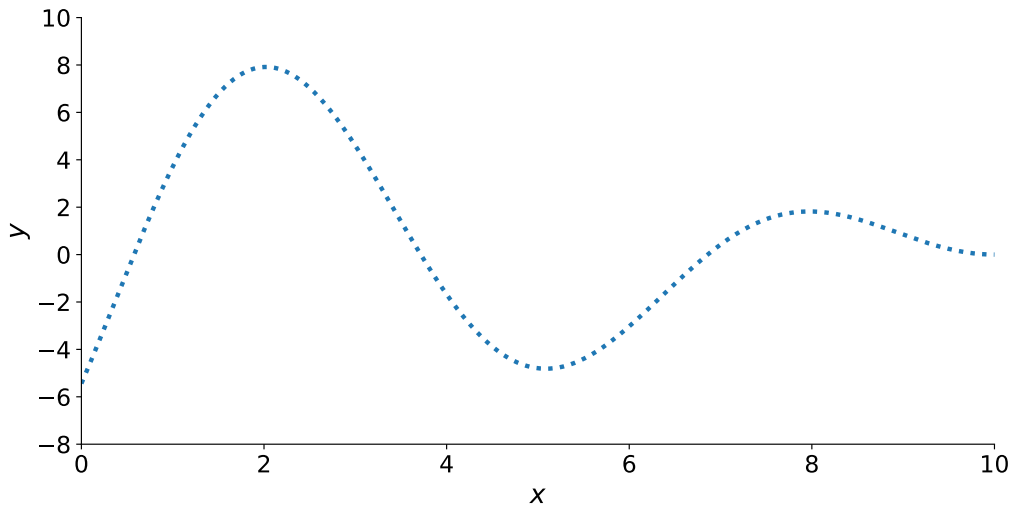


BME513 - Inteligencia Artificial en Salud

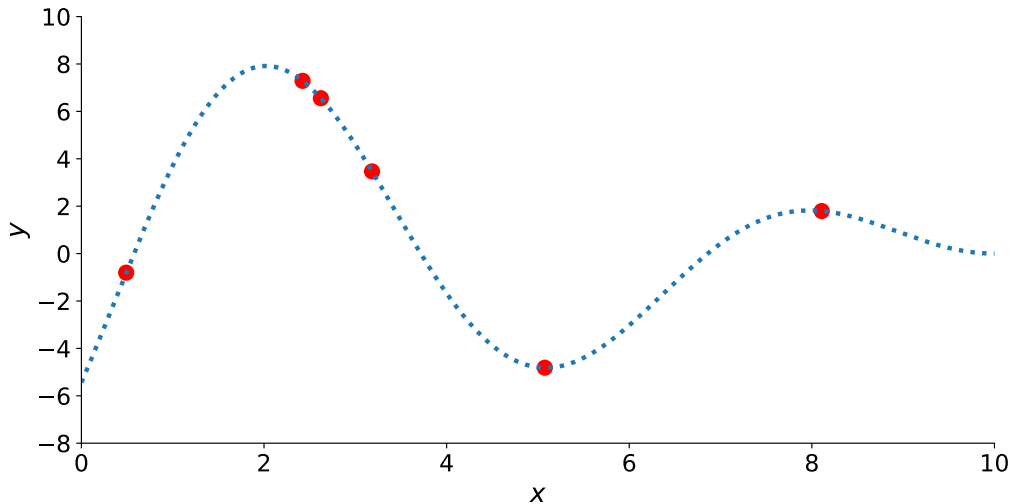
Gaussian processes

Alejandro Veloz

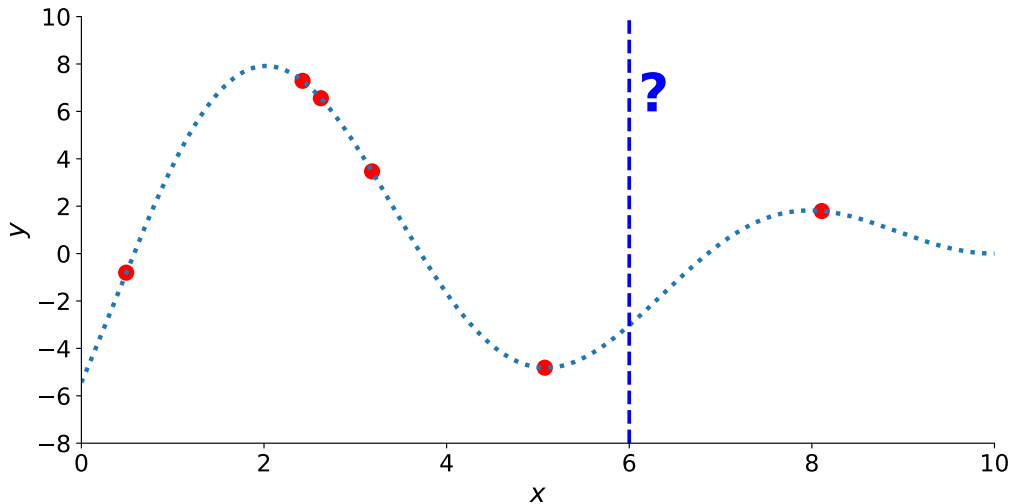
Motivation: non-linear regression



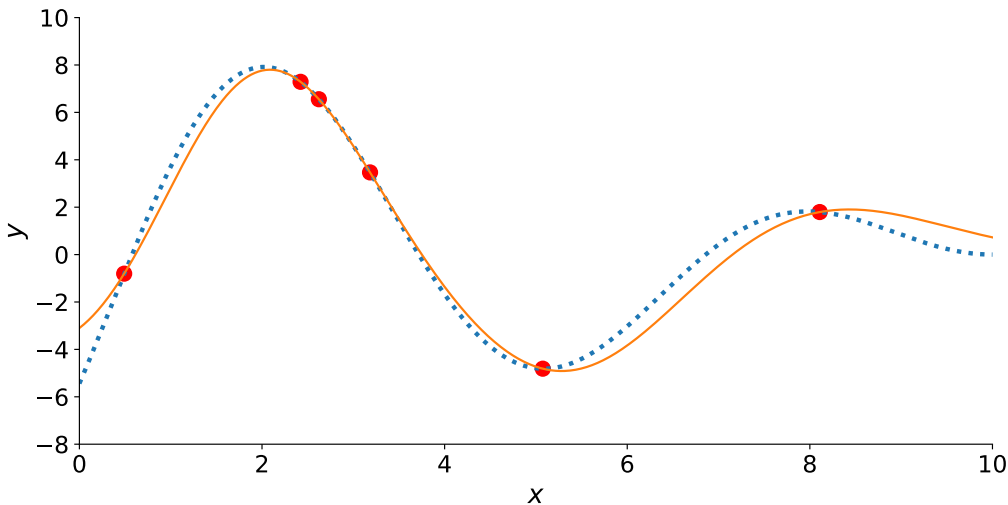
Motivation: non-linear regression



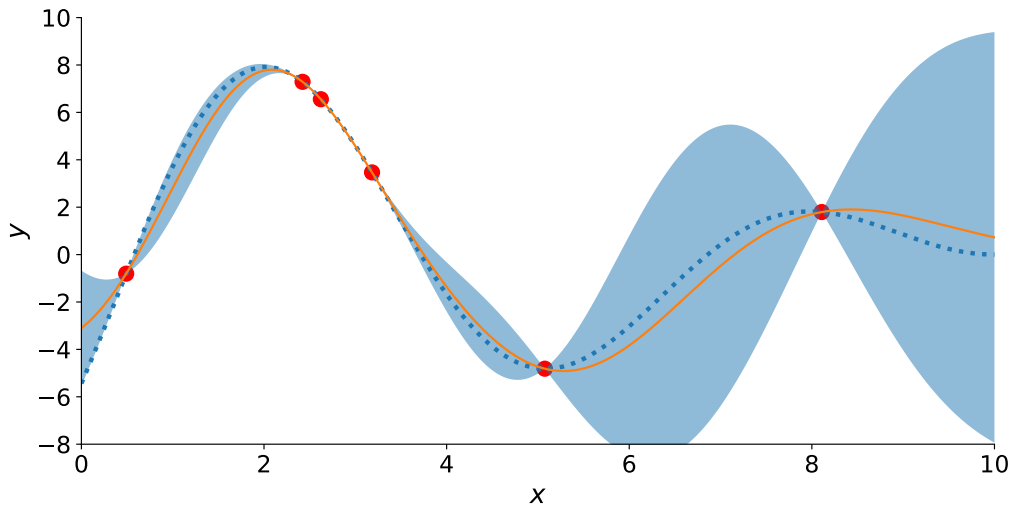
Motivation: non-linear regression



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Outline

From gaussian distribution to Gaussian processes

Gaussian processes for regression

Covariance functions

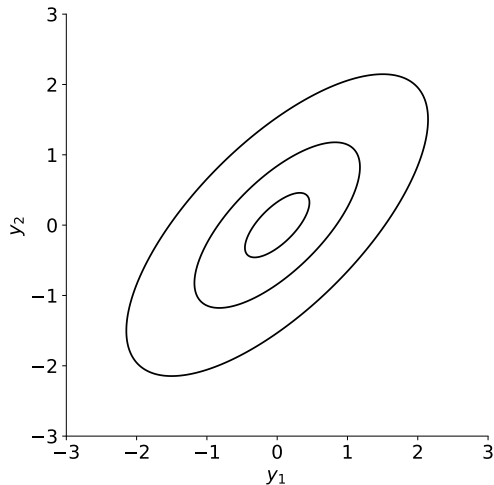
Applications

Resources

Gaussian distribution

$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

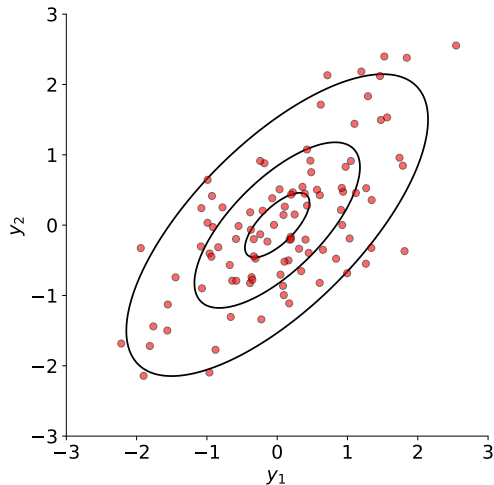
$$\Sigma = \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix}$$



Gaussian distribution

$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

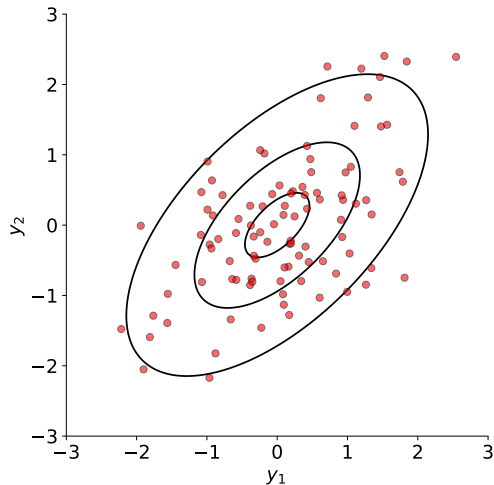
$$\Sigma = \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix}$$



Gaussian distribution

$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

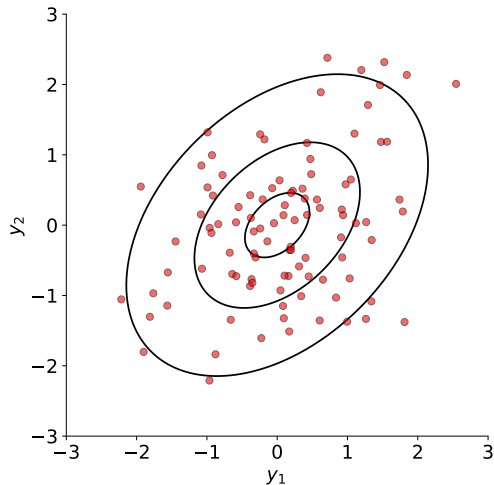
$$\Sigma = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix}$$



Gaussian distribution

$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

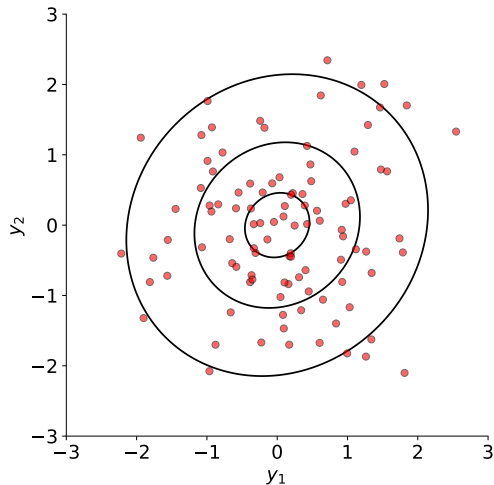
$$\Sigma = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix}$$



Gaussian distribution

$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

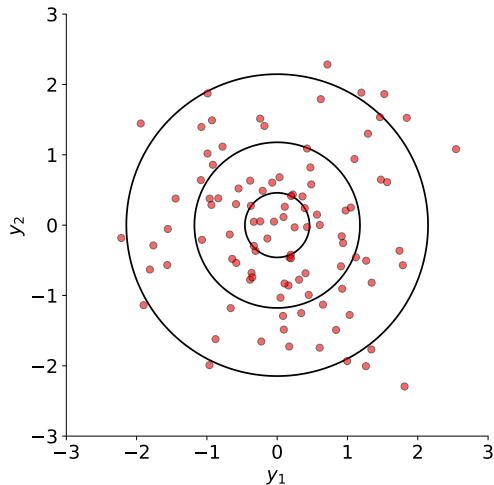
$$\Sigma = \begin{bmatrix} 1 & 0.1 \\ 0.1 & 1 \end{bmatrix}$$



Gaussian distribution

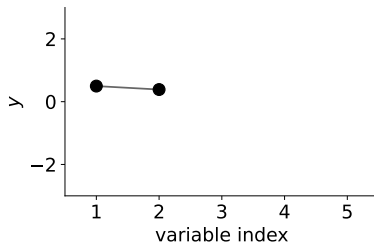
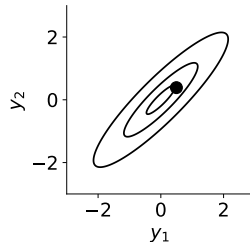
$$p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^\top \Sigma^{-1} \mathbf{y}\right)$$

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



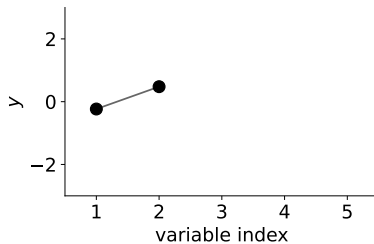
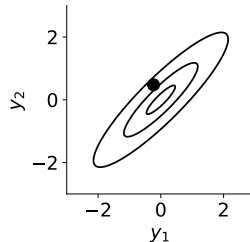
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



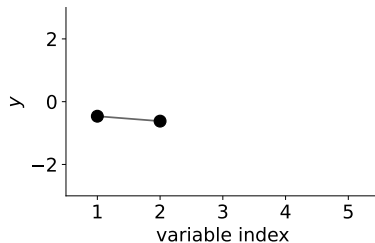
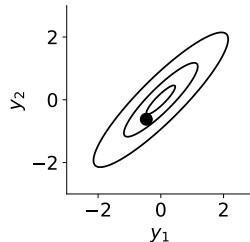
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



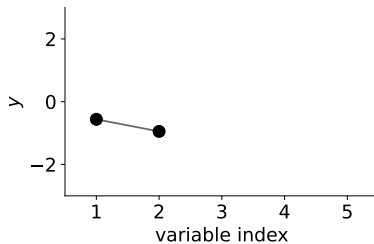
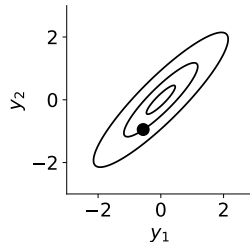
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



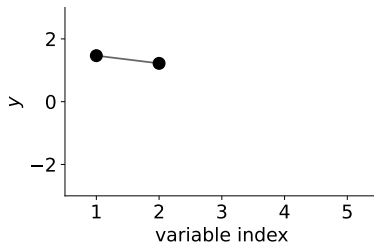
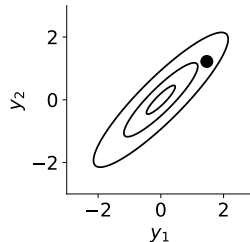
New visualization

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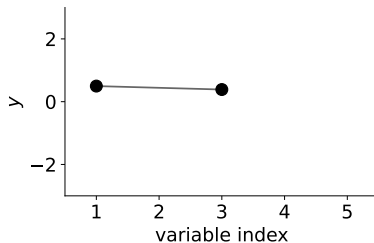
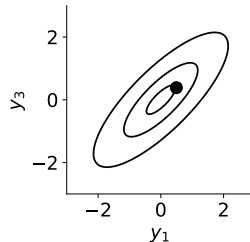
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



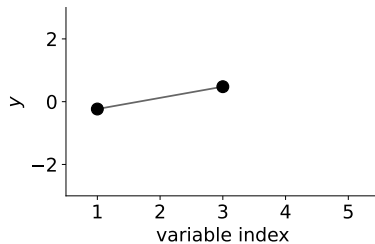
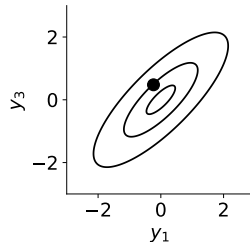
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



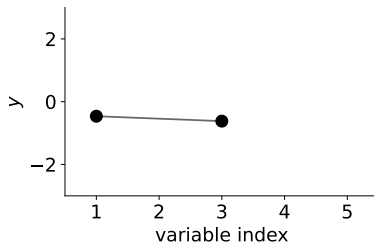
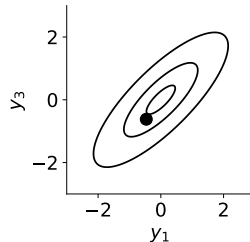
New visualization

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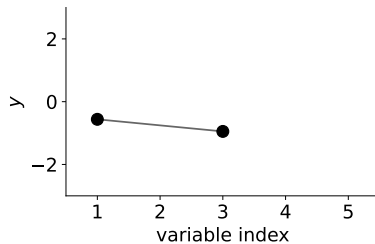
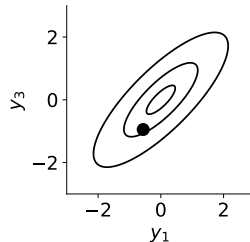
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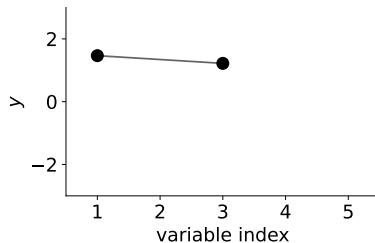
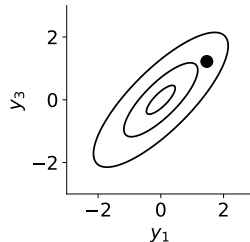
New visualization

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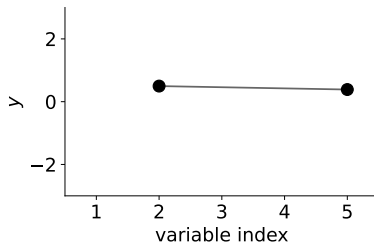
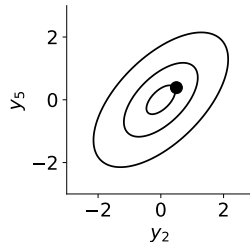
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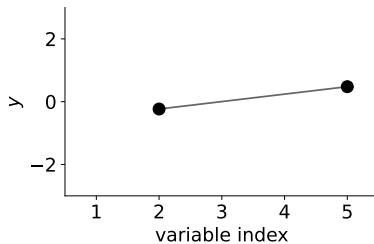
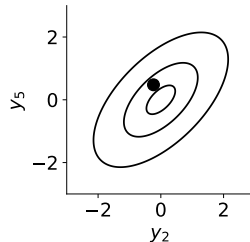
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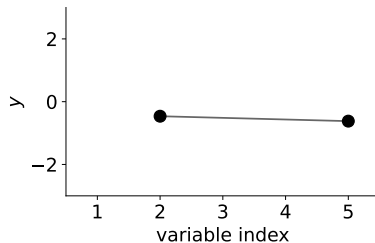
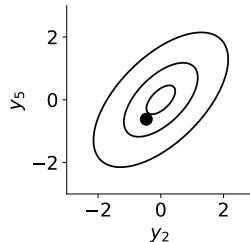
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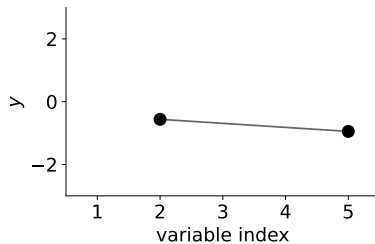
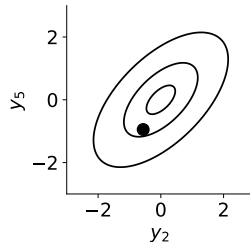
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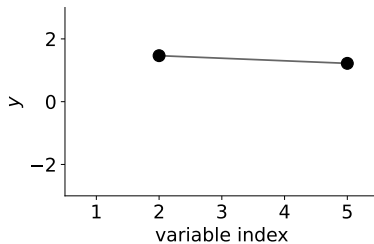
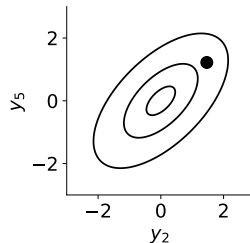
New visualization

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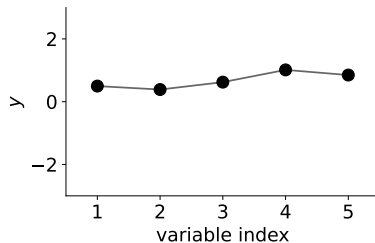
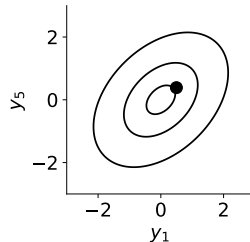
New visualization

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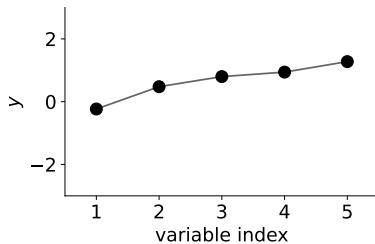
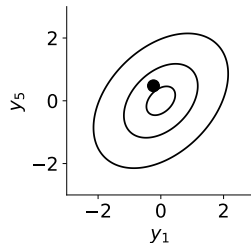
New visualization

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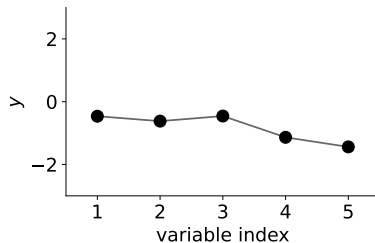
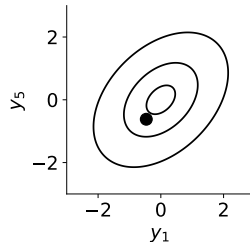
New visualization

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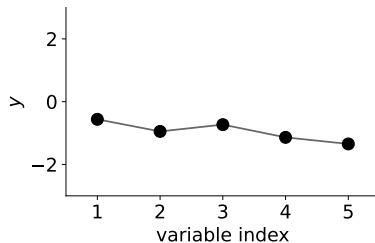
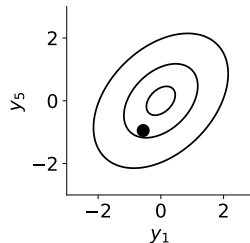
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.8 & 0.6 & 0.4 \\ 0.9 & 1 & 0.9 & 0.8 & 0.6 \\ 0.8 & 0.9 & 1 & 0.9 & 0.8 \\ 0.6 & 0.8 & 0.9 & 1 & 0.9 \\ 0.4 & 0.6 & 0.8 & 0.9 & 1 \end{bmatrix}$$



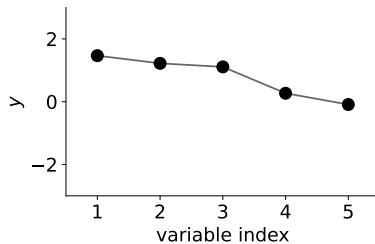
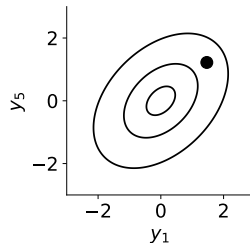
New visualization

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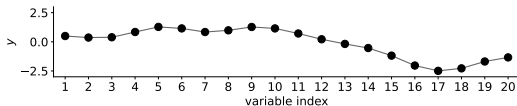
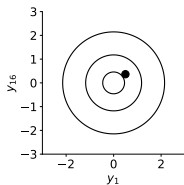
New visualization

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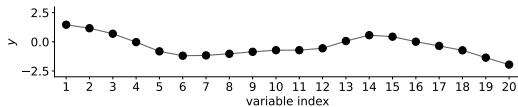
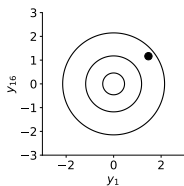
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & 0 & \dots \\ 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & \dots \\ 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & \dots \\ 0.3 & 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$



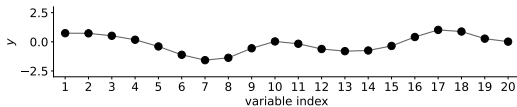
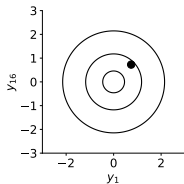
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & 0 & \dots \\ 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & \dots \\ 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & \dots \\ 0.3 & 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$



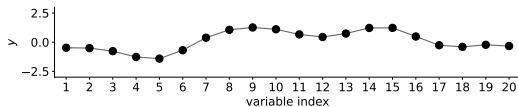
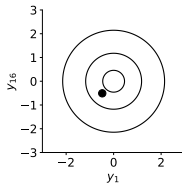
New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & 0 & \dots \\ 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & \dots \\ 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & \dots \\ 0.3 & 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$



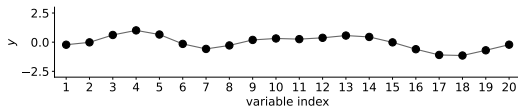
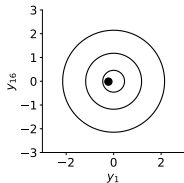
New visualization

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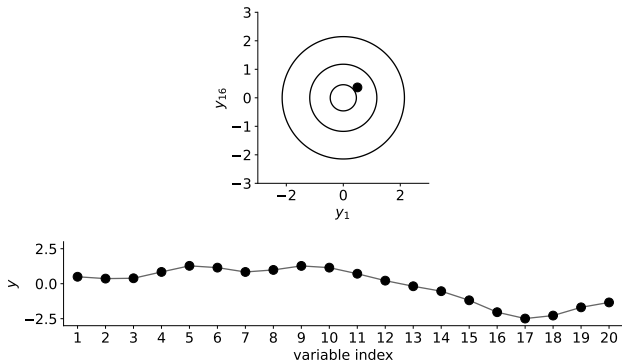
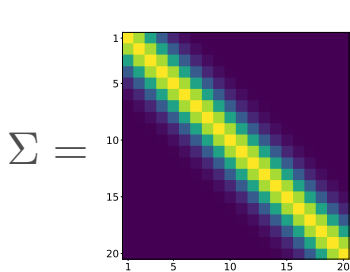


New visualization

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & 0 & \dots \\ 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & 0 & \dots \\ 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & 0 & \dots \\ 0.3 & 0.6 & 0.9 & 1 & 0.9 & 0.6 & 0.3 & 0.1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

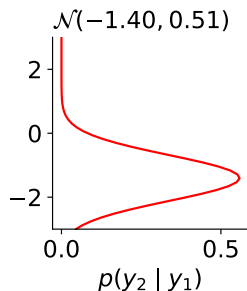
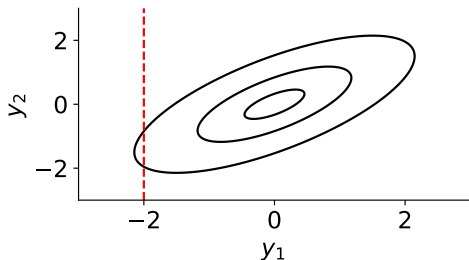
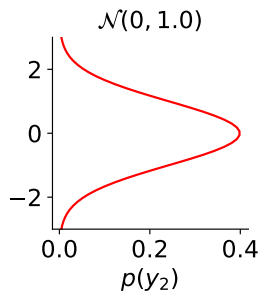


New visualization



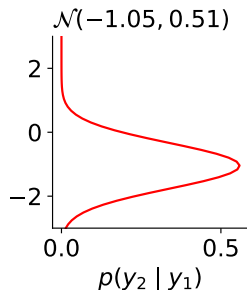
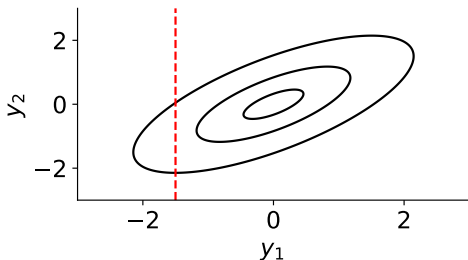
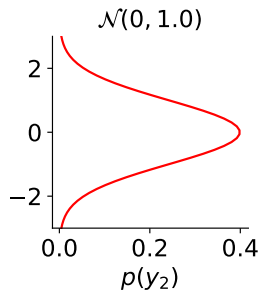
Marginalization and conditioning

$$p(\mathbf{y} \mid \Sigma) \propto \exp \left(-\frac{1}{2} \mathbf{y}^\top \Sigma^{-1} \mathbf{y} \right) \quad \Sigma = \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix}$$



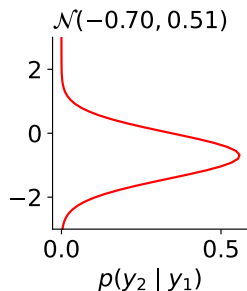
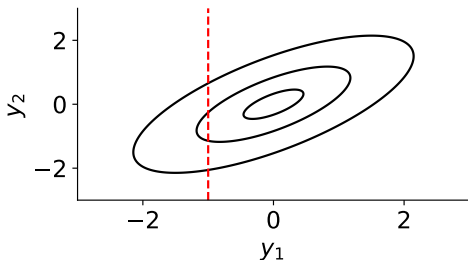
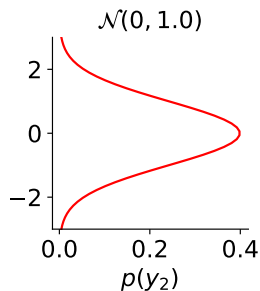
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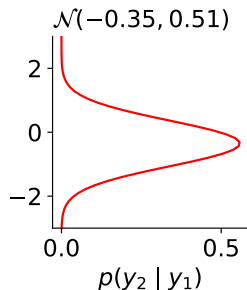
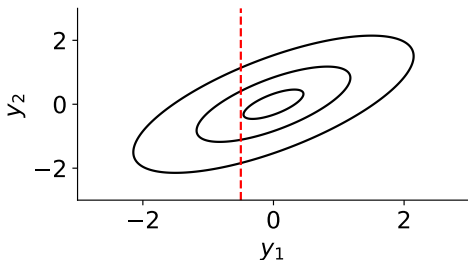
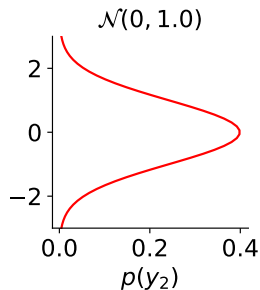
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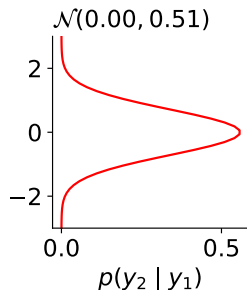
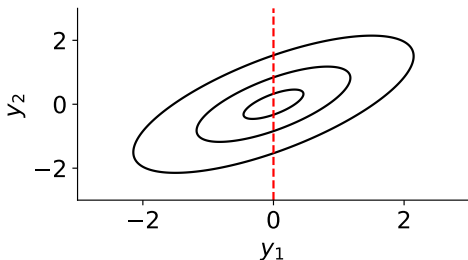
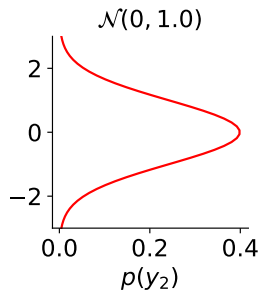
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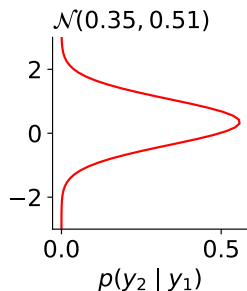
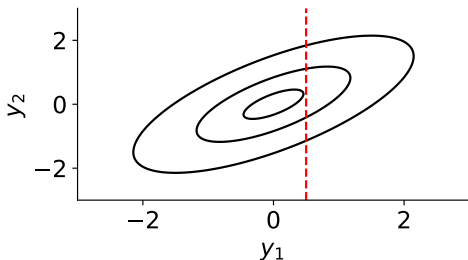
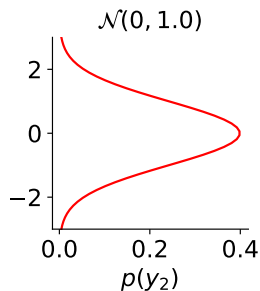
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$$p(\mathbf{y} \mid \Sigma) \propto \exp \left(-\frac{1}{2} \mathbf{y}^T \Sigma^{-1} \mathbf{y} \right) \quad \Sigma = \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix}$$



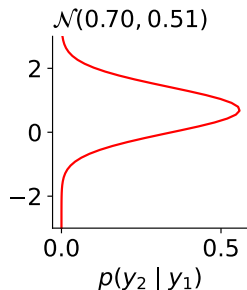
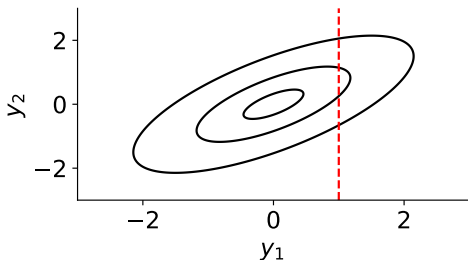
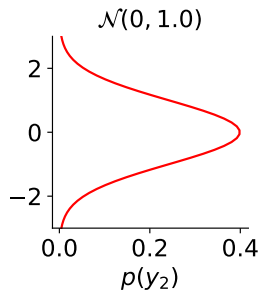
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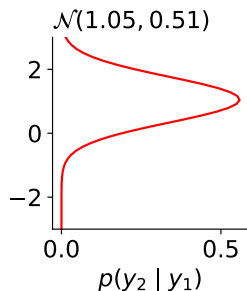
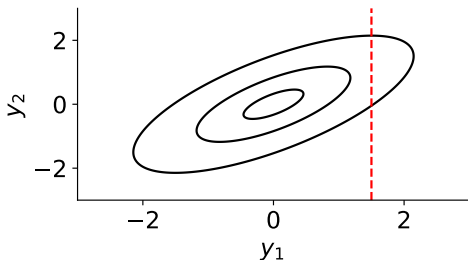
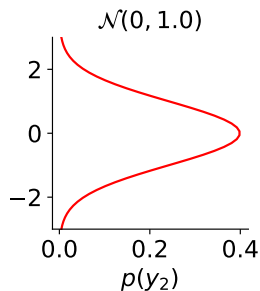
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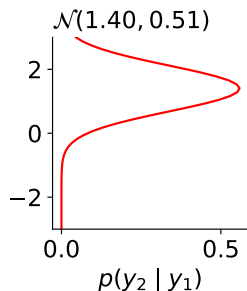
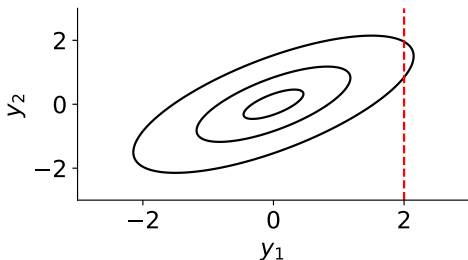
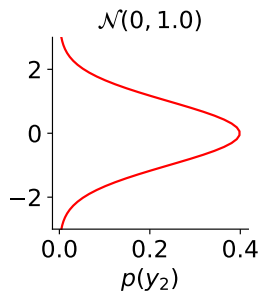
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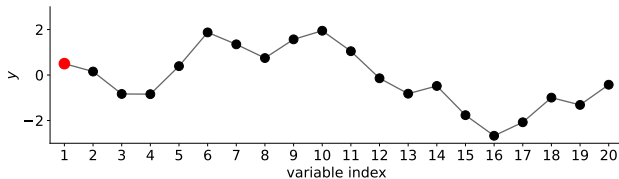
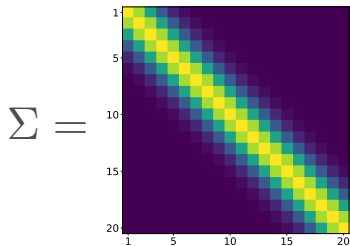


Marginalization and conditioning

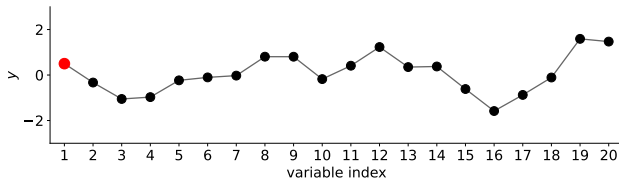
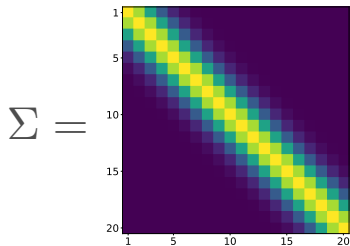
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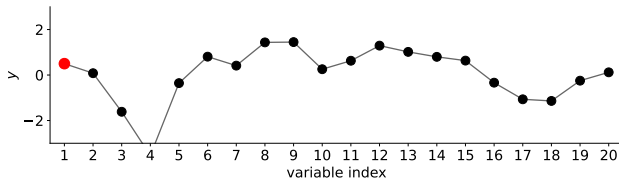
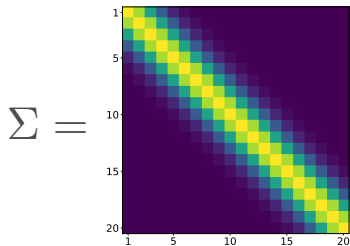
Marginalization and conditioning



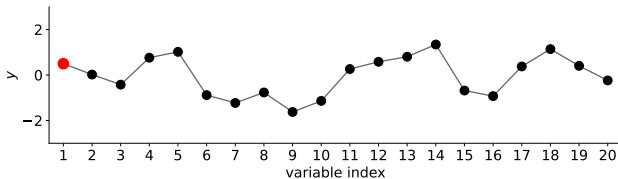
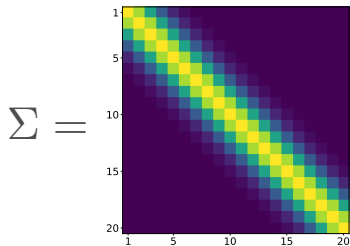
Marginalization and conditioning



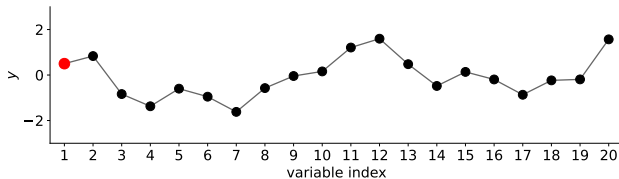
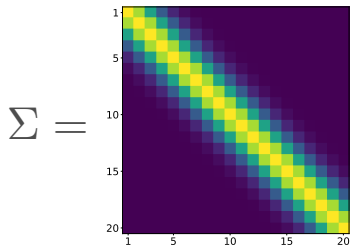
Marginalization and conditioning



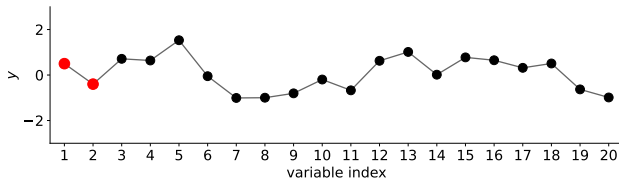
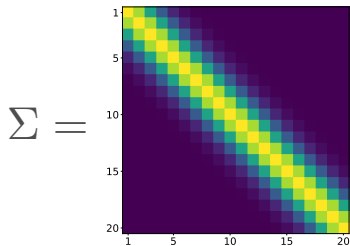
Marginalization and conditioning



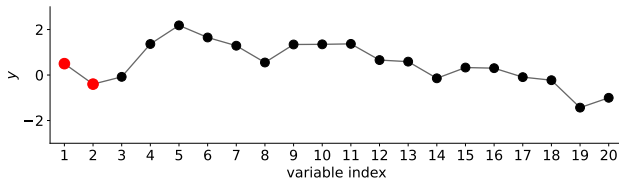
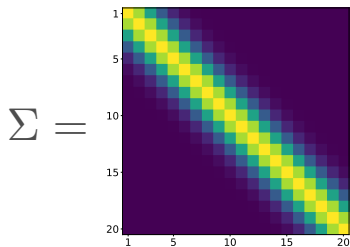
Marginalization and conditioning



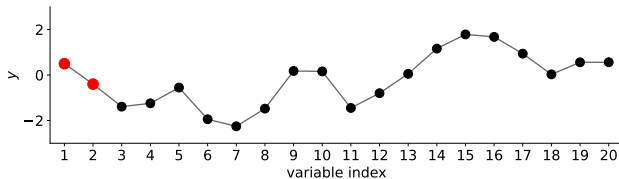
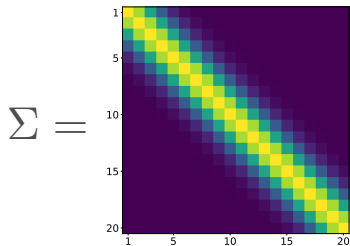
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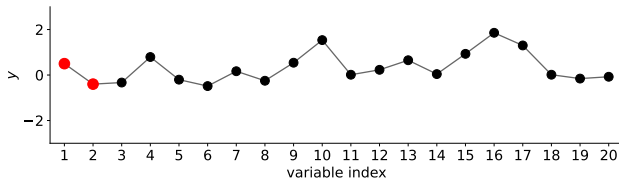
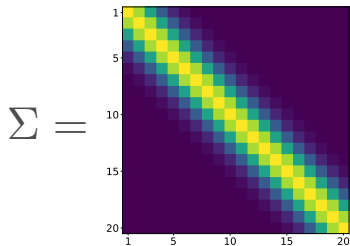
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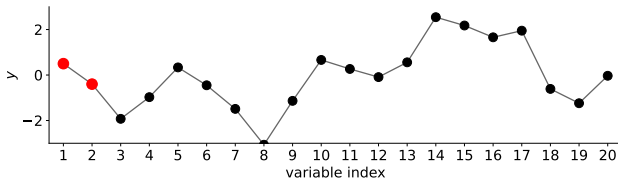
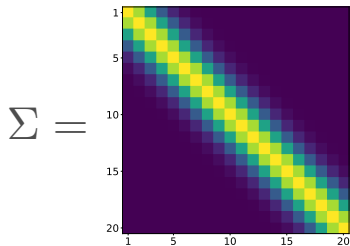
Marginalization and conditioning



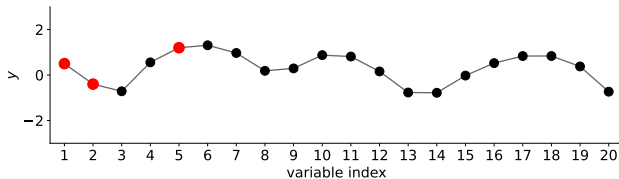
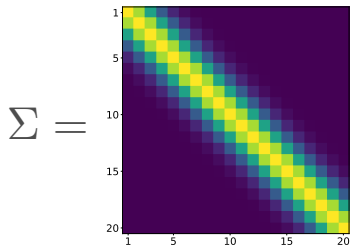
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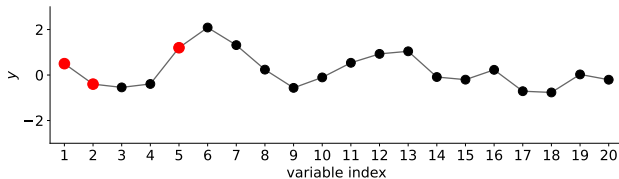
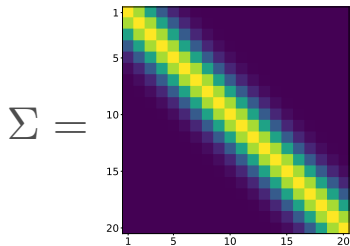
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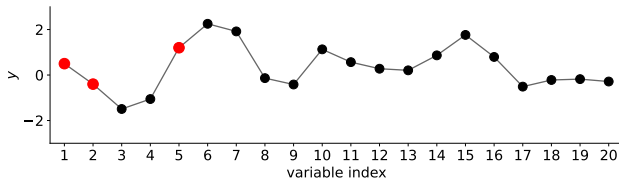
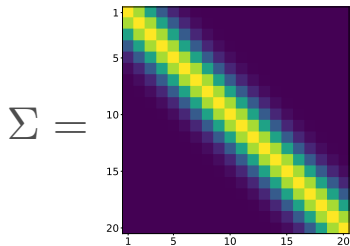
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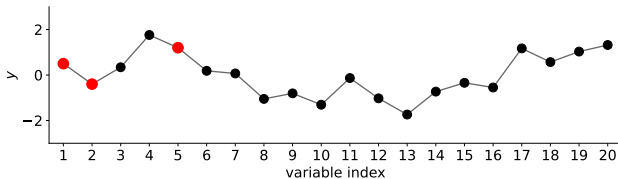
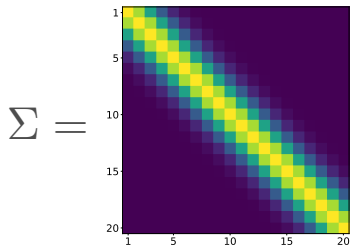
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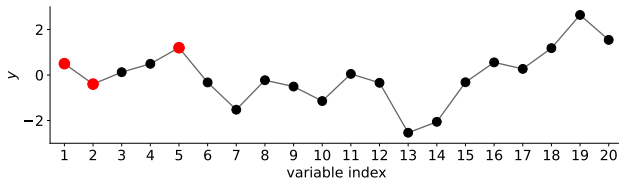
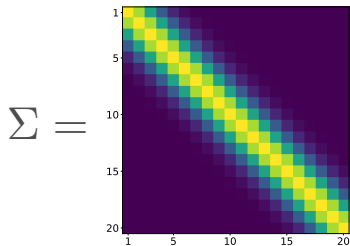
Marginalization and conditioning



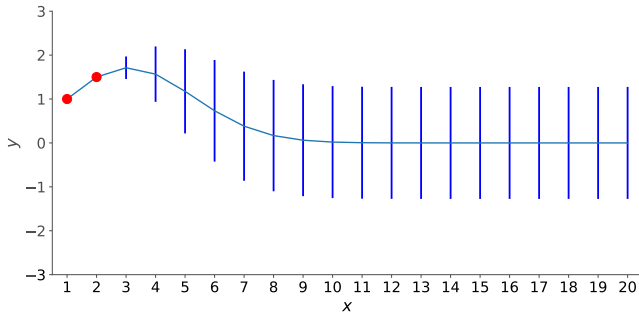
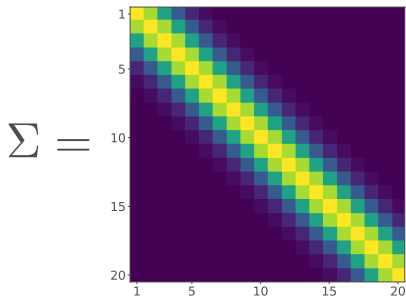
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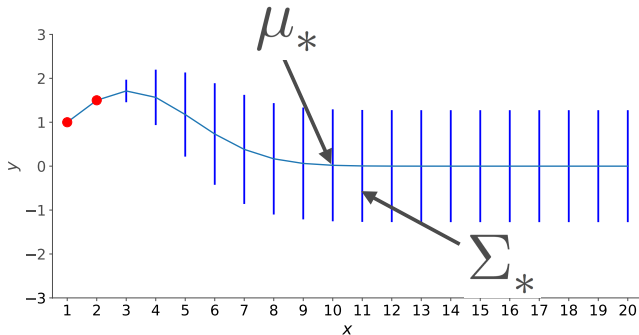
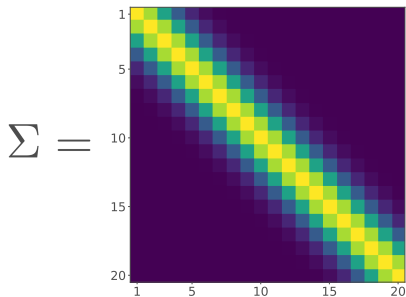
Marginalization and conditioning



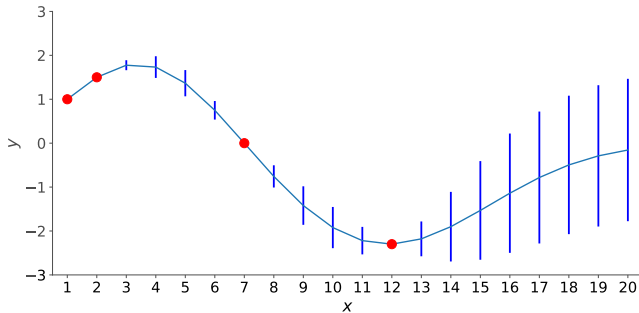
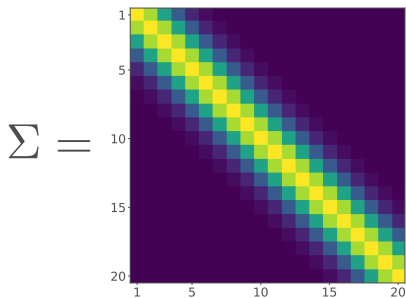
Regression using Gaussians



Regression using Gaussians

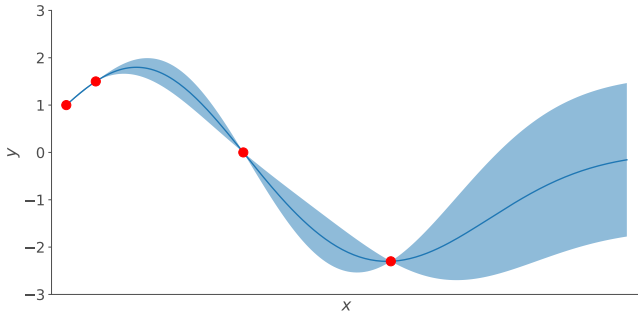
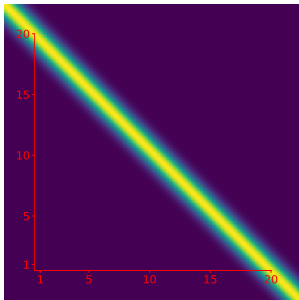


Regression using Gaussians



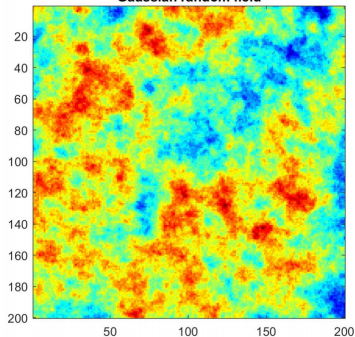
Regression using Gaussians

$\Sigma =$

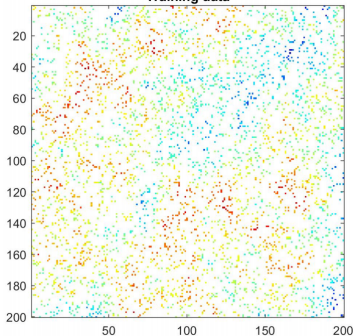


Regression using Gaussians

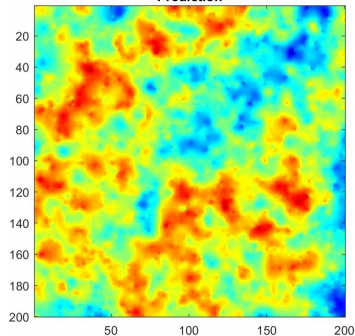
Gaussian random field



Training data



Prediction



Gaussian processes in machine learning

- Gaussian processes (GPs), also known as Gaussian random fields.
- Originating in geostatistics, they were introduced by George Matheron around 1960 under the name kriging.
- Their use in machine learning has grown substantially since the 1990s.
- A Gaussian process extends the multivariate Gaussian distribution to an infinite collection of random variables indexed by input points.
- Its most common use in machine learning is probabilistic non-linear regression.
- Gaussian processes are also applied in pattern classification, dimensionality reduction, missing-data, multi-task learning, and Bayesian optimization.

Gaussian process

For any finite set of points, this process defines a joint Gaussian:

$$p(\mathbf{f} \mid \mathbf{X}) = \mathcal{N}(\mathbf{f} \mid \boldsymbol{\mu}, \mathbf{K})$$

where $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ and $\boldsymbol{\mu} = (m(\mathbf{x}_1), \dots, m(\mathbf{x}_N))$.

Gaussian Process (GP)

A Gaussian Process (GP) is denoted by:

$$f(\mathbf{x}) \sim GP(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

where $m(\mathbf{x})$ is the **mean function** and $k(\mathbf{x}, \mathbf{x}')$ is the **kernel** or **covariance function**.

Mean and covariance function

To fully specify the law of a GP, we need to specify mean and covariance functions:

$$f(\mathbf{x}_1), \dots, f(\mathbf{x}_N) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad \forall \mathbf{x}, \mathbf{x}' \in \mathcal{X} \subset \mathbb{R}^d$$

where

mean function

$$m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$$

[kernel covariance function]

$$\begin{aligned} k(\mathbf{x}, \mathbf{x}') &= \text{Cov}[f(\mathbf{x}), f(\mathbf{x}')] \\ &= \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))] \end{aligned}$$

Mean function $m(\mathbf{x})$

We can use any mean function we want $m(x) = \mathbb{E}[f(\mathbf{x})]$.

Popular choices are:

- $m(\mathbf{x}) = 0$
- $m(\mathbf{x}) = \text{const}$
- $m(x) = \phi(\mathbf{x})^\top \mathbf{w}$
- Neural networks.

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Popular choices are:

Let $f(\mathbf{x}) = \phi(\mathbf{x})^\top \mathbf{w}$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I})$.

Mean function $m(\mathbf{x})$

We can use any mean function we want $m(x) = \mathbb{E}[f(\mathbf{x})]$.

Popular choices are:

- $m(\mathbf{x}) = 0$
- $m(\mathbf{x}) = \text{const}$
- $m(x) = \phi(\mathbf{x})^\top \mathbf{w}$
- Neural networks.

Let $f(\mathbf{x}) = \phi(\mathbf{x})^\top \mathbf{w}$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I})$.

Then $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$

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Kernel covariance function $k(\mathbf{x}, \mathbf{x}')$

The covariance function determines the nature of the GP, i.e., the hypothesis space/space of functions.

We usually use a covariance function that is a function of the indexes/locations:

$$k(\mathbf{x}, \mathbf{x}') = \text{Cov}(f(\mathbf{x}), f(\mathbf{x}')) ,$$

$k(\cdot, \cdot)$ must satisfy:

- Symmetry, i.e., $k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}', \mathbf{x})$.
- For any locations $\mathbf{x}_1, \dots, \mathbf{x}_N$, the $N \times N$ Gram matrix $\mathbf{K}$ with $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ must be a positive semi-definite matrix.

Kernel covariance function $k(\mathbf{x}, \mathbf{x}')$

Let $f(\mathbf{x}) = \phi(\mathbf{x})^\top \mathbf{w}$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I})$.

The kernel covariance $k(\mathbf{x}, \mathbf{x}')$ is given by:

$$\begin{aligned} k(\mathbf{x}, \mathbf{x}') &= \mathbb{E}[f(\mathbf{x})f(\mathbf{x}')] \\ &= \phi(\mathbf{x})^\top \mathbb{E}[\mathbf{w}\mathbf{w}^\top] \phi(\mathbf{x}') \\ &= \phi(\mathbf{x})^\top \frac{\mathbf{I}}{\alpha} \phi(\mathbf{x}') \\ &= \frac{\phi(\mathbf{x})^\top \phi(\mathbf{x}')}{\alpha} \end{aligned}$$

Kernel covariance function $k(\mathbf{x}, \mathbf{x}')$

A widely used kernel in GPs is the squared-exponential or RBF (radial basis function) kernel is given by:

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right), \quad x \in \mathbb{R}$$

where σ_f^2 is the signal variance parameter and ℓ the length-scale parameter.

How do we draw samples from a GP?

- Given the mean function and covariance function for a GP, we can draw samples using a multivariate Gaussian distribution.
- To sample from the multivariate Gaussian distribution, we need a mean vector and a covariance matrix.
- The mean vector is obtained from the mean function.
- The covariance matrix is obtained from the covariance function.

Sampling from a GP, example

We have the set of x values:

$$\{x_1, x_2, \dots, x_N\} \subseteq \mathbb{R}$$

These are the indexes of the stochastic process.

We now compute the covariance matrix

$$\mathbf{K} = \begin{bmatrix} k(x_1, x_1) & k(x_1, x_2) & \cdots & k(x_1, x_N) \\ k(x_2, x_1) & k(x_2, x_2) & \cdots & k(x_2, x_N) \\ \vdots & \vdots & \vdots & \vdots \\ k(x_N, x_1) & k(x_N, x_2) & \cdots & k(x_N, x_N) \end{bmatrix}$$

We assume the mean function is constant and equal to zero, i.e.,
 $m(x) = 0 \forall x$.

Sampling from a GP, example

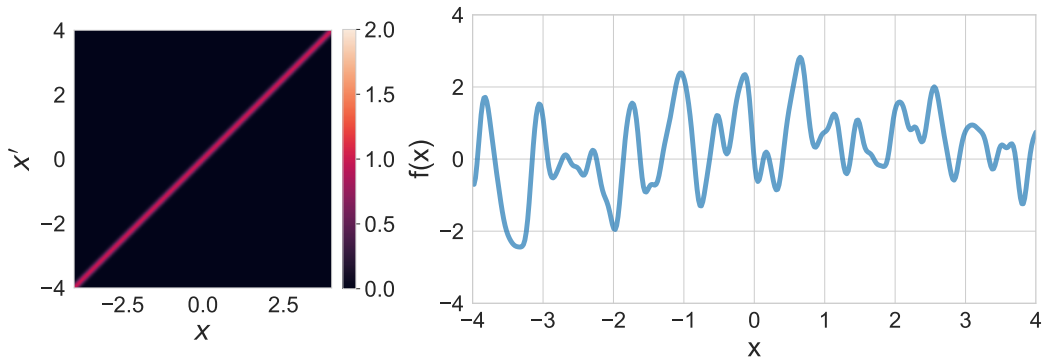
To generate functions from this GP, we will then sample from:

$$f(x_1), \dots, f(x_N) \\ \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & k(x_1, x_2) & \cdots & k(x_1, x_N) \\ k(x_2, x_1) & k(x_2, x_2) & \cdots & k(x_2, x_N) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_N, x_1) & k(x_N, x_2) & \cdots & k(x_N, x_N) \end{bmatrix} \right)$$

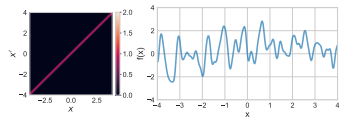
What we plot is x_i and $f(x_i)$, $\forall i$.

Sampling from a GP, example

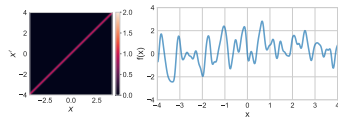
$$\ell = 0.1, \sigma_f = 1$$



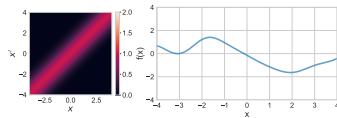
$$\ell = 0.1, \sigma_f = 1$$



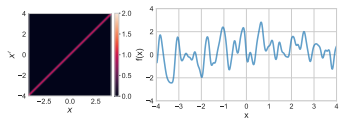
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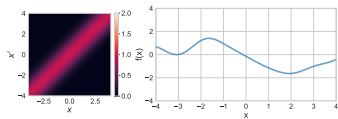
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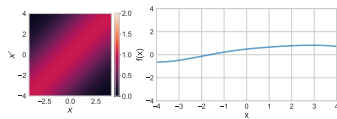
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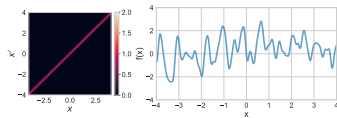
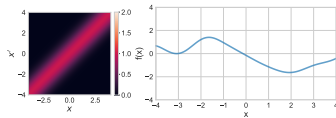
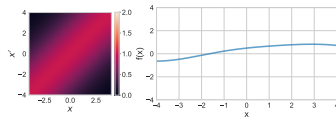
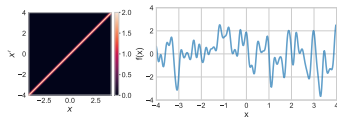


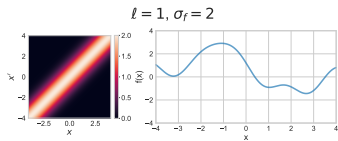
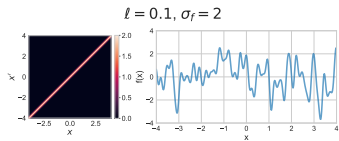
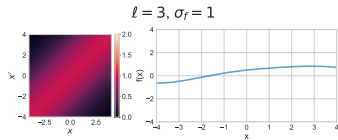
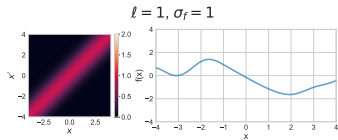
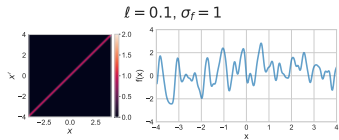
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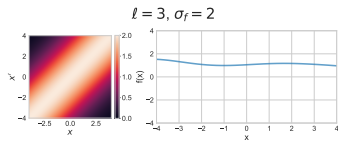
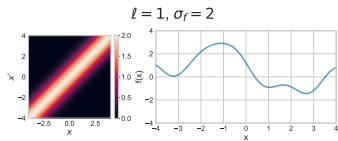
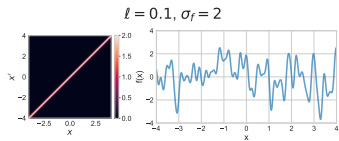
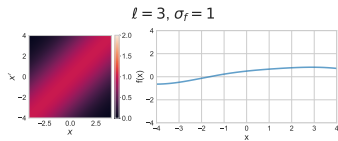
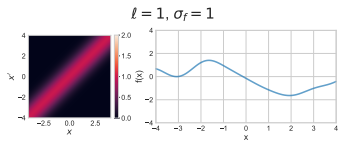
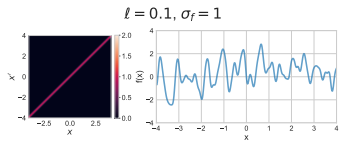


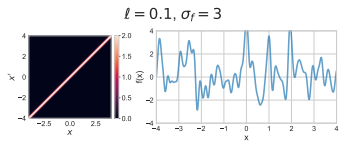
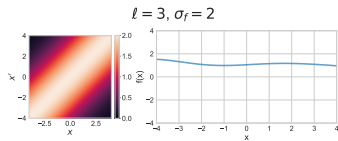
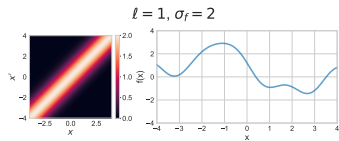
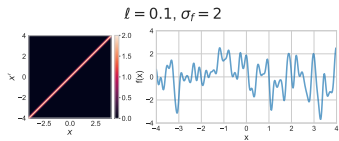
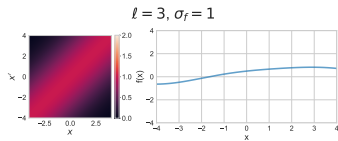
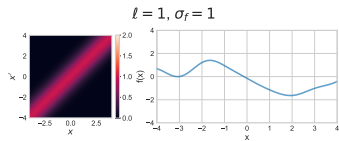
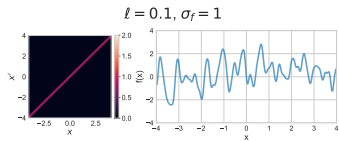
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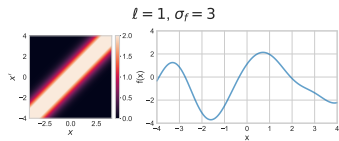
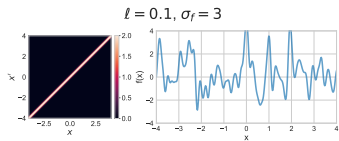
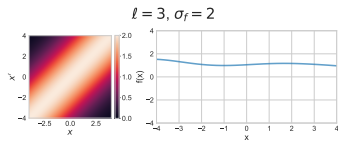
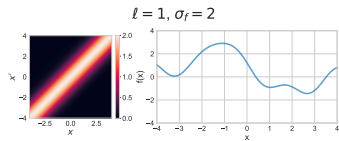
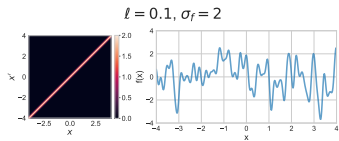
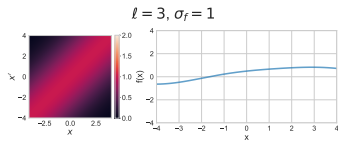
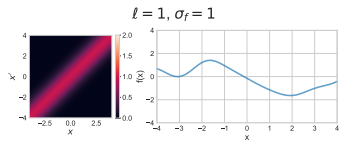
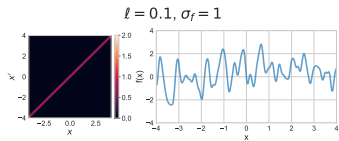


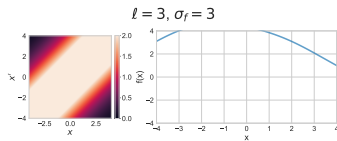
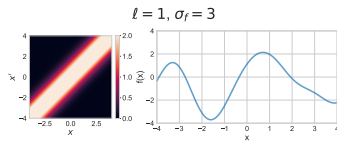
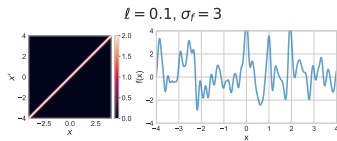
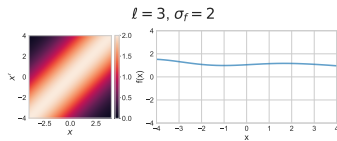
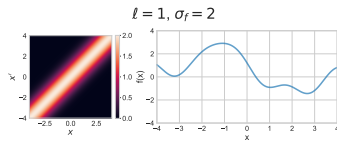
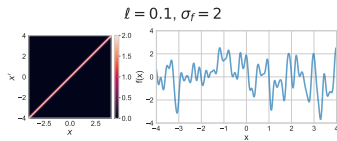
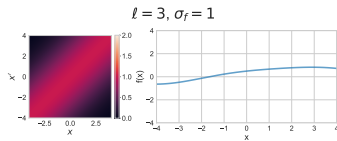
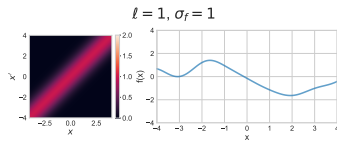
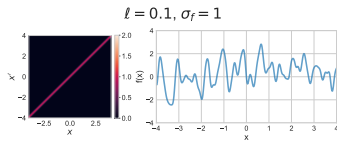
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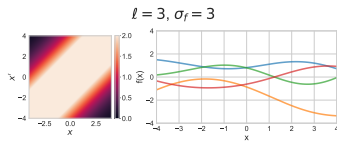
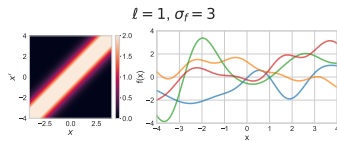
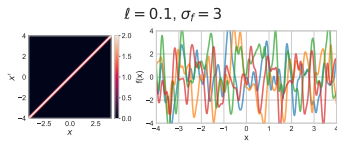
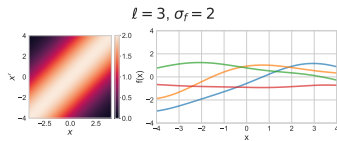
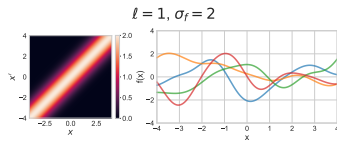
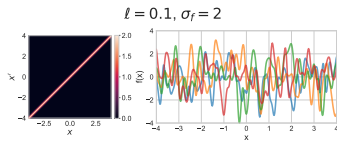
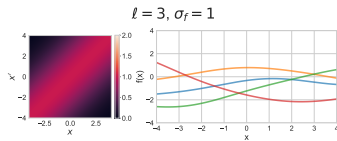
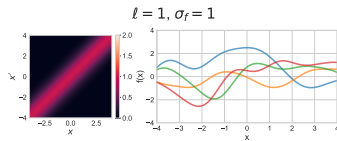
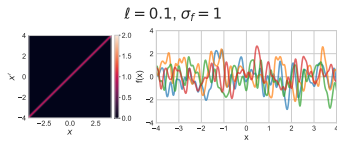






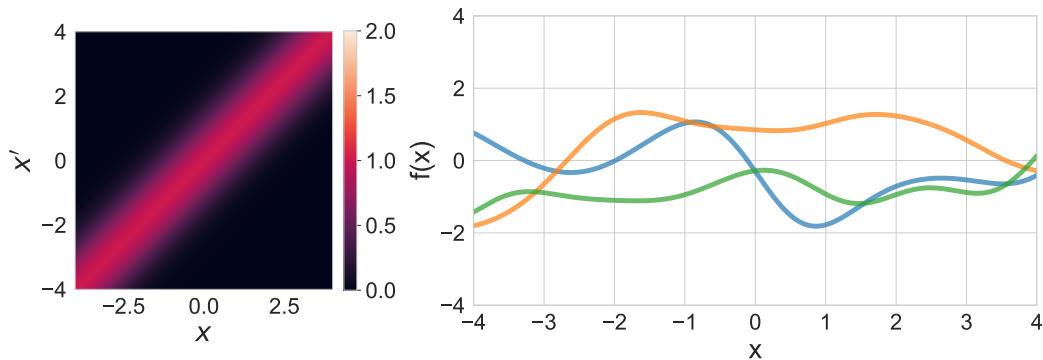






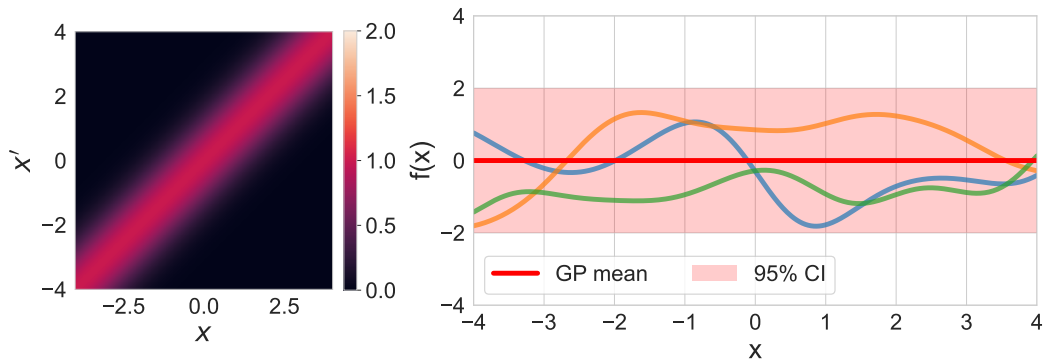
From prior to posterior

$$\ell = 1, \sigma_f = 1$$



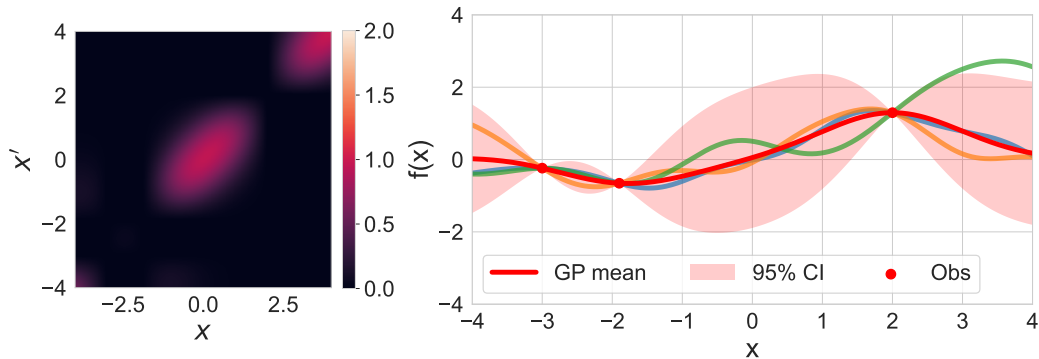
From prior to posterior

$$\ell = 1, \sigma_f = 1$$



From prior to posterior

$$\ell = 1, \sigma_f = 1$$



From prior to posterior

Training set:

$$\mathcal{D} = \{(\mathbf{x}_i, f_i), i = 1 : N\}$$

Test set:

$$\{\mathbf{x}_i^*, i = 1 : N_*\}$$

The training and test sets are organized as follows:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix}$$

training inputs

We want to predict the function outputs $\mathbf{f}^*$.

From prior to posterior

Training set:

$$\mathcal{D} = \{(\mathbf{x}_i, f_i), i = 1 : N\}$$

Test set:

$$\{\mathbf{x}_i^*, i = 1 : N_*\}$$

The training and test sets are organized as follows:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} \quad \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix}$$

training inputs training targets

We want to predict the function outputs $\mathbf{f}^*$.

From prior to posterior

Training set:

$$\mathcal{D} = \{(\mathbf{x}_i, f_i), i = 1 : N\}$$

Test set:

$$\{\mathbf{x}_i^*, i = 1 : N_*\}$$

The training and test sets are organized as follows:

$$\begin{array}{ccc} \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} & \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix} & \mathbf{X}^* = \begin{pmatrix} \mathbf{x}_1^{*\top} \\ \vdots \\ \mathbf{x}_{N_*}^{*\top} \end{pmatrix} \\ \text{training inputs} & \text{training targets} & \text{test inputs} \end{array}$$

We want to predict the function outputs $\mathbf{f}^*$.

From prior to posterior

Training set:

$$\mathcal{D} = \{(\mathbf{x}_i, f_i), i = 1 : N\}$$

Test set:

$$\{\mathbf{x}_i^*, i = 1 : N_*\}$$

The training and test sets are organized as follows:

$$\begin{array}{ccccc} \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} & \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix} & \mathbf{X}^* = \begin{pmatrix} \mathbf{x}_1^{*\top} \\ \vdots \\ \mathbf{x}_{N_*}^{*\top} \end{pmatrix} & \mathbf{f}^* = \begin{pmatrix} f_1^* \\ \vdots \\ f_{N_*}^* \end{pmatrix} \\ \text{training inputs} & \text{training targets} & \text{test inputs} & \text{predictions} \end{array}$$

We want to predict the function outputs $\mathbf{f}^*$.

From prior to posterior

The prior GP joint distribution of $\mathbf{f}$ and $\mathbf{f}^*$ is multivariate normal, i.e.:

$$\begin{pmatrix} \mathbf{f} \\ \mathbf{f}_* \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \boldsymbol{\mu} \\ \boldsymbol{\mu}_* \end{pmatrix}, \begin{pmatrix} \mathbf{K} & \mathbf{K}_* \\ \mathbf{K}_*^\top & \mathbf{K}_{**} \end{pmatrix} \right)$$

where:

$\mathbf{K}$ is of size $N \times N$, with $[\mathbf{K}]_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.

$\mathbf{K}_*$ is of size $N \times N_*$, with $[\mathbf{K}_*]_{ij} = k(\mathbf{x}_i, \mathbf{x}_{*j})$.

$\mathbf{K}_{**}$ is of size $N_* \times N_*$, with $[\mathbf{K}_{**}]_{ij} = k(\mathbf{x}_{*i}, \mathbf{x}_{*j})$.

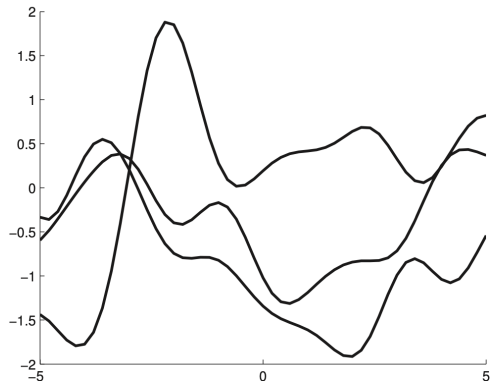
From prior to posterior

Conditioning on the training targets $\mathbf{f}$ gives the posterior GP, i.e.:

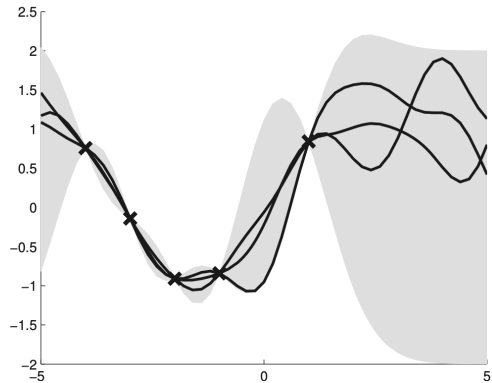
$$\mathbf{f}_* \mid \mathbf{X}_*, \mathbf{X}, \mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*)$$

The posterior has the following form:

$$\begin{aligned}\boldsymbol{\mu}_* &= \boldsymbol{\mu}(\mathbf{X}_*) + \mathbf{K}_*^\top \mathbf{K}^{-1}(\mathbf{f} - \boldsymbol{\mu}) \\ \boldsymbol{\Sigma}_* &= \mathbf{K}_{**} - \mathbf{K}_*^\top \mathbf{K}^{-1} \mathbf{K}_*\end{aligned}$$



(a)

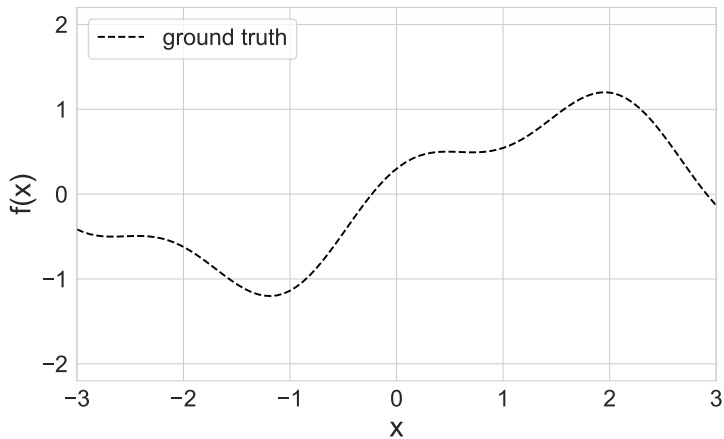


(b)

(a) Samples from the prior, $p(\mathbf{f} \mid \mathbf{X})$, using a squared exponential kernel.

(b) Samples from a GP posterior, $p(\mathbf{f}_* \mid \mathbf{X}_*, \mathbf{X}, \mathbf{f})$.

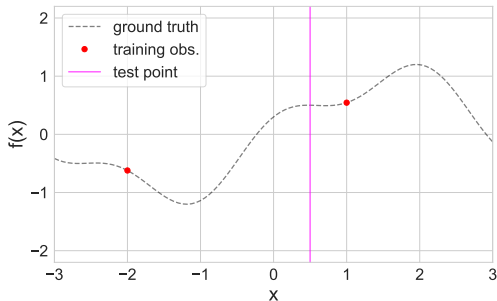
Numerical example



Numerical example



Numerical example



Training samples:

$$(x_1, f(x_1)) = (-2, -0.62)$$

$$(x_2, f(x_2)) = (1, 0.54)$$

$$x^* = 0.5$$

$$f^* = ?$$

Numerical example

Suppose f is a Gaussian process, then

$$\underbrace{f(x_1), f(x_2)}_{\text{training samples}}, \underbrace{f(x^*)}_{\text{test sample}} \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

where:

$$\Sigma = \left(\begin{array}{cc|c} k(x_1, x_1) & k(x_1, x_2) & k(x_1, x^*) \\ k(x_2, x_1) & k(x_2, x_2) & k(x_2, x^*) \\ \hline k(x^*, x_1) & k(x^*, x_2) & k(x^*, x^*) \end{array} \right) = \left(\begin{array}{cc|c} 1 & 0.01 & 0.04 \\ 0.01 & 1 & 0.88 \\ \hline 0.04 & 0.88 & 1 \end{array} \right)$$

The value of the function f^* at the testing point $x^* = 0.5$ is computed from $p(\mathbf{f}_* | \mathbf{f}) = \mathcal{N}(\boldsymbol{\mu}_*, *)$ with:

$$\begin{aligned} \boldsymbol{\mu}_* &= \mathbf{K}_*^\top \mathbf{K}^{-1} (\mathbf{f} - \boldsymbol{\mu}) \\ &= \begin{pmatrix} k(x^*, x_1) & k(x^*, x_2) \end{pmatrix} \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) \\ k(x_2, x_1) & k(x_2, x_2) \end{pmatrix}^{-1} \begin{pmatrix} f(x_1) \\ f(x_2) \end{pmatrix} \\ &= 0.46 \end{aligned}$$

Numerical example

Suppose f is a Gaussian process, then

$$\underbrace{f(x_1), f(x_2)}_{\text{training samples}}, \underbrace{f(x^*)}_{\text{test sample}} \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

where:

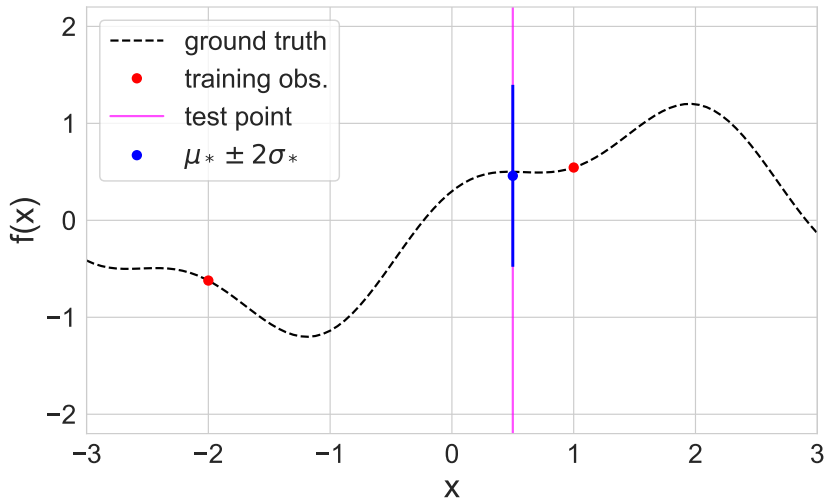
$$\Sigma = \left(\begin{array}{cc|c} k(x_1, x_1) & k(x_1, x_2) & k(x_1, x^*) \\ k(x_2, x_1) & k(x_2, x_2) & k(x_2, x^*) \\ \hline k(x^*, x_1) & k(x^*, x_2) & k(x^*, x^*) \end{array} \right) = \left(\begin{array}{cc|c} 1 & 0.01 & 0.04 \\ 0.01 & 1 & 0.88 \\ \hline 0.04 & 0.88 & 1 \end{array} \right)$$

The value of the function f^* at the testing point $x^* = 0.5$ is computed from $p(\mathbf{f}_* | \mathbf{f}) = \mathcal{N}(\boldsymbol{\mu}_*, *)$ with:

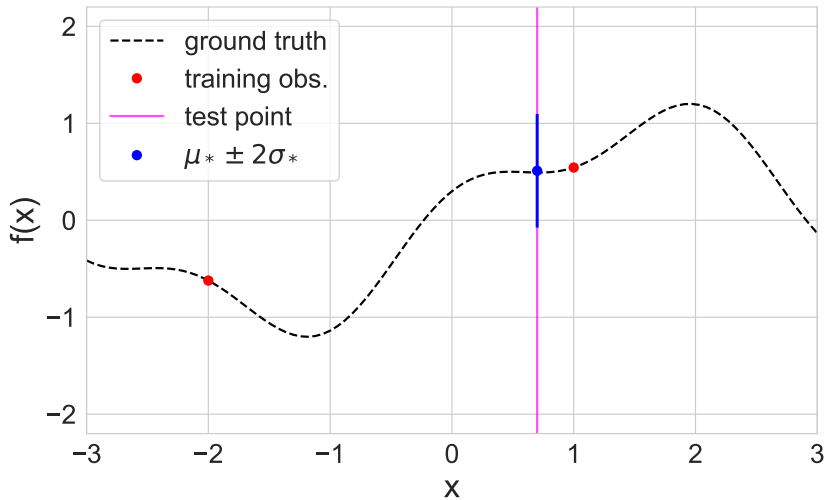
$$\Sigma_* = \mathbf{K}_{**} - \mathbf{K}_*^T \mathbf{K}^{-1} \mathbf{K}_*$$

$$\begin{aligned} &= k(x^*, x^*) - \begin{pmatrix} k(x^*, x_1) & k(x^*, x_2) \end{pmatrix} \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) \\ k(x_2, x_1) & k(x_2, x_2) \end{pmatrix}^{-1} \begin{pmatrix} k(x^*, x_1) \\ k(x^*, x_2) \end{pmatrix} \\ &= 0.22 \end{aligned}$$

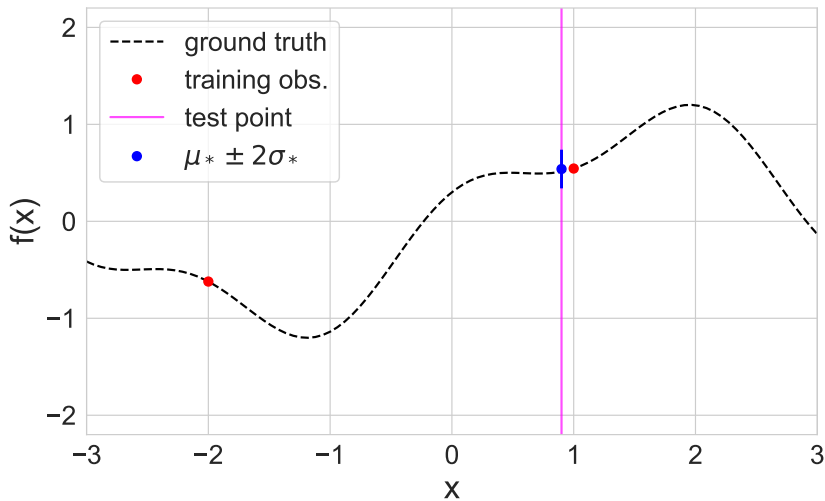
Numerical example



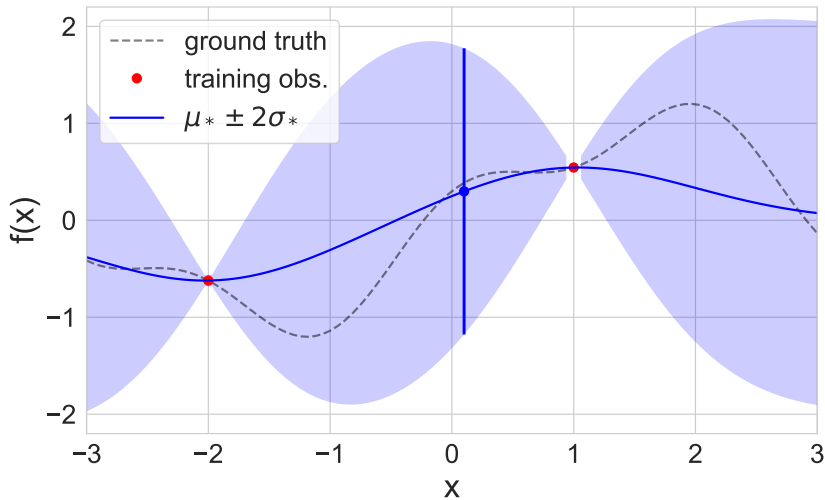
Numerical example



Numerical example



Numerical example



Predictions using noisy observations

Noisy version of the underlying function:

$$y = f(\mathbf{x}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_y^2)$$

The covariance of the observed noisy responses is:

$$\text{cov}[y_p, y_q] = K(\mathbf{x}_p, \mathbf{x}_q) + \sigma_y^2 \mathbb{I}(p = q)$$

$$\text{cov}[\mathbf{y} \mid \mathbf{X}] = \mathbf{K} + \sigma_y^2 \mathbf{I}_N \triangleq \mathbf{K}_y$$

Predictions using noisy observations

The joint density of training and test samples is given by:

$$y_1, y_2, \dots, y_N, f_* \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \mathbf{K}_y & \mathbf{k}_* \\ \mathbf{k}_*^\top & k_{**} \end{pmatrix} \right)$$

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{f}_* \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \mathbf{K}_y & \mathbf{K}_* \\ \mathbf{K}_*^\top & \mathbf{K}_{**} \end{pmatrix} \right)$$

Predictions using noisy observations

The posterior predictive density is:

$$\begin{aligned}p(\mathbf{f}_* \mid \mathbf{X}_*, \mathbf{X}, \mathbf{y}) &= \mathcal{N}(\mathbf{f}_* \mid \boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*) \\ \boldsymbol{\mu}_* &= \mathbf{K}_*^T \mathbf{K}_y^{-1} \mathbf{y} \\ \boldsymbol{\Sigma}_* &= \mathbf{K}_{**} - \mathbf{K}_*^T \mathbf{K}_y^{-1} \mathbf{K}_*\end{aligned}$$

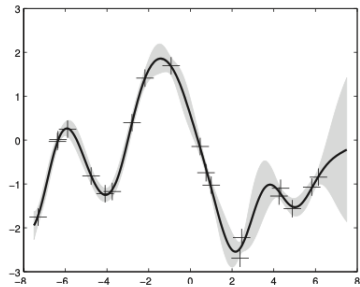
In the case of a single test input, this simplifies as follows:

$$p(f_* \mid \mathbf{x}_*, \mathbf{X}, \mathbf{y}) = \mathcal{N}(f_* \mid \mathbf{k}_*^T \mathbf{K}_y^{-1} \mathbf{y}, k_{**} - \mathbf{k}_*^T \mathbf{K}_y^{-1} \mathbf{k}_*)$$

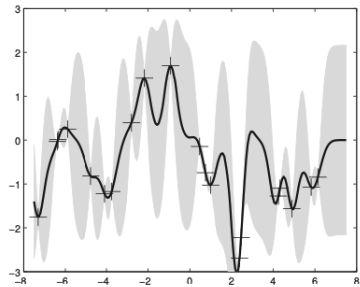
where $\mathbf{k}_* = [\kappa(\mathbf{x}_*, \mathbf{x}_1), \dots, \kappa(\mathbf{x}_*, \mathbf{x}_N)]$ and $k_{**} = \kappa(\mathbf{x}_*, \mathbf{x}_*)$. Another way to write the posterior mean is as follows:

$$\bar{f}_* = \mathbf{k}_*^T \mathbf{K}_y^{-1} \mathbf{y} = \sum_{i=1}^N \alpha_i \kappa(\mathbf{x}_i, \mathbf{x}_*)$$

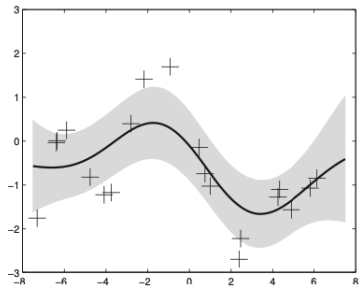
where $\boldsymbol{\alpha} = \mathbf{K}_y^{-1} \mathbf{y}$.



(a)



(b)



(c)

Which kernel functions are appropriate for a given problem?

How does the choice of different hyperparameter values affect the resulting model?

Kernels in Gaussian Processes

A **kernel** (or covariance function) $k(\mathbf{x}, \mathbf{x}')$ is the foundation of a Gaussian process. It defines the covariance structure between function values at any two input points, completely specifying the GP's prior distribution and determining:

- **Smoothness** of sample functions
- **Length-scale** (characteristic distance over which correlations decay)
- **Amplitude** (overall vertical scale of variations)
- **Periodicity** (if applicable)
- **Non-stationarity** (if applicable)

Kernels must be **positive semi-definite** to ensure valid covariance matrices.

Kernel Effects

1. **Signal Variance** (σ_f^2)

- Controls the **amplitude** or overall vertical scale of function variations.
- Larger values: function can vary over a wider range.
- Smaller values: function values stay closer to the mean.

2. **Length-Scale** (ℓ)

- Controls how quickly correlations **decay with distance**.
- **Large** ℓ : Points remain strongly correlated over longer distances. Sample functions are smooth and slowly varying.
- **Small** ℓ : Correlations decay rapidly, function can vary more rapidly and rougher behavior.

Two common kernels

RBF (squared exponential) kernel:

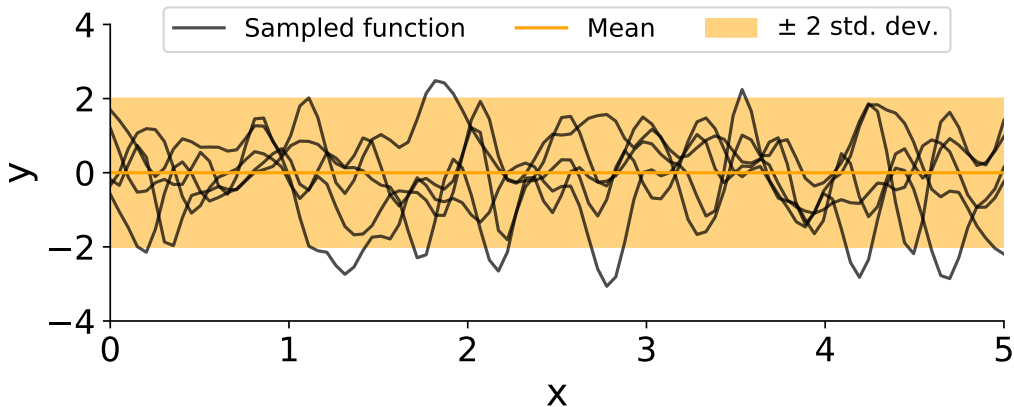
$$k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right).$$

Matérn kernel (general $\nu > 0$) can be written as:

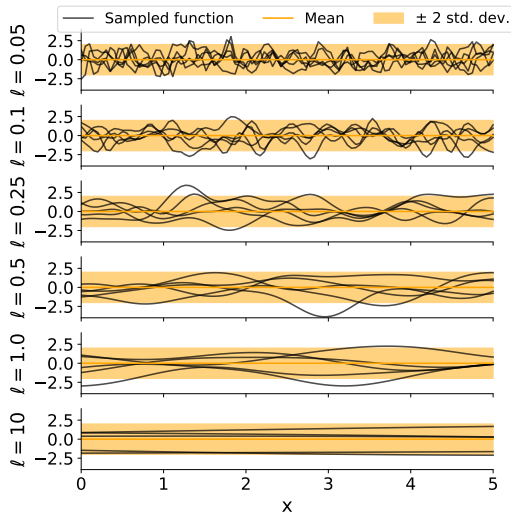
$$k_{\text{Matérn}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \frac{1}{\Gamma(\nu) 2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{\ell} \|\mathbf{x} - \mathbf{x}'\|\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}}{\ell} \|\mathbf{x} - \mathbf{x}'\|\right),$$

where K_ν is the modified Bessel function of the second kind, ν controls smoothness, ℓ is the length scale, and σ_f^2 is the signal variance.

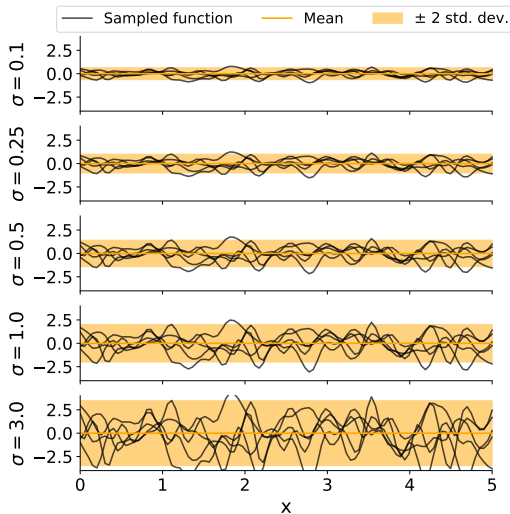
RBF kernel - samples from prior



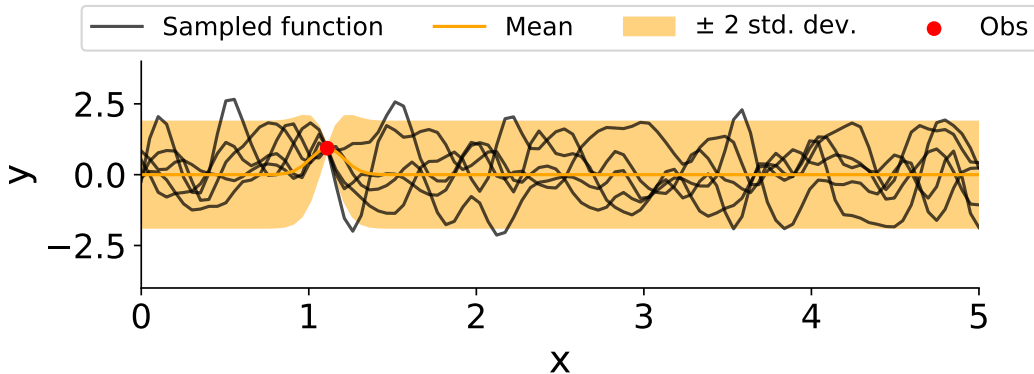
RBF kernel - samples from prior



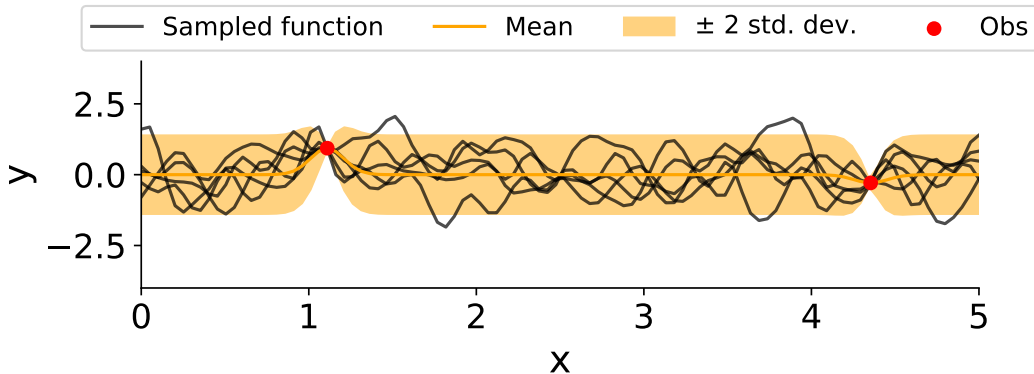
RBF kernel - samples from prior



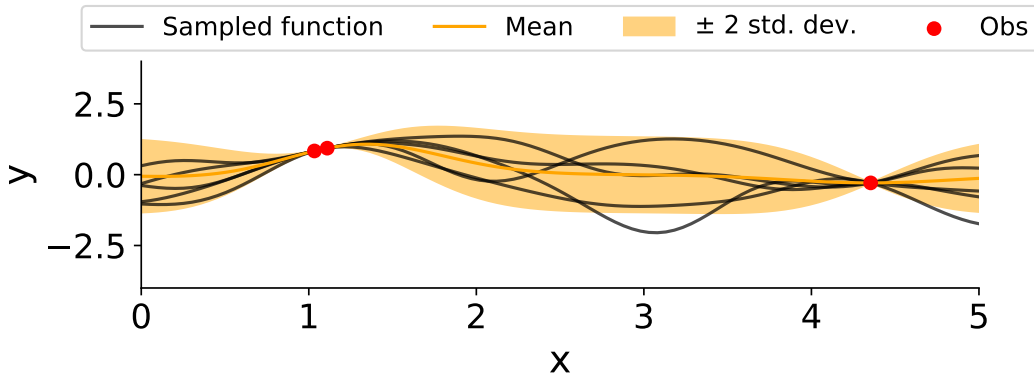
RBF kernel - samples from posterior



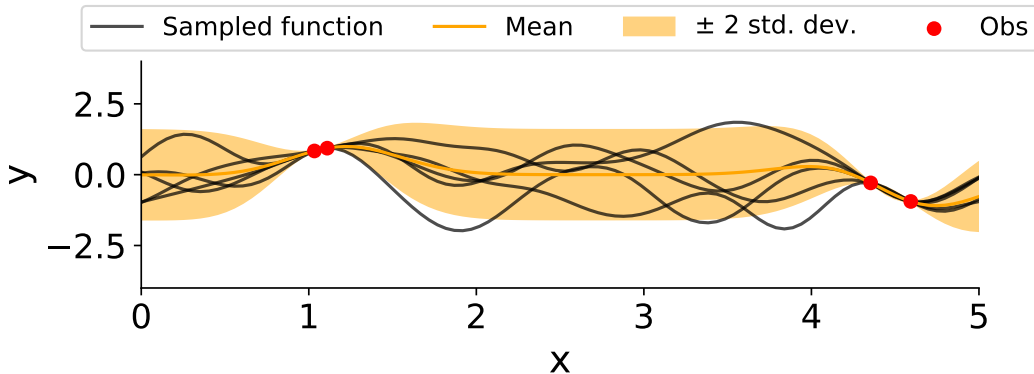
RBF kernel - samples from posterior



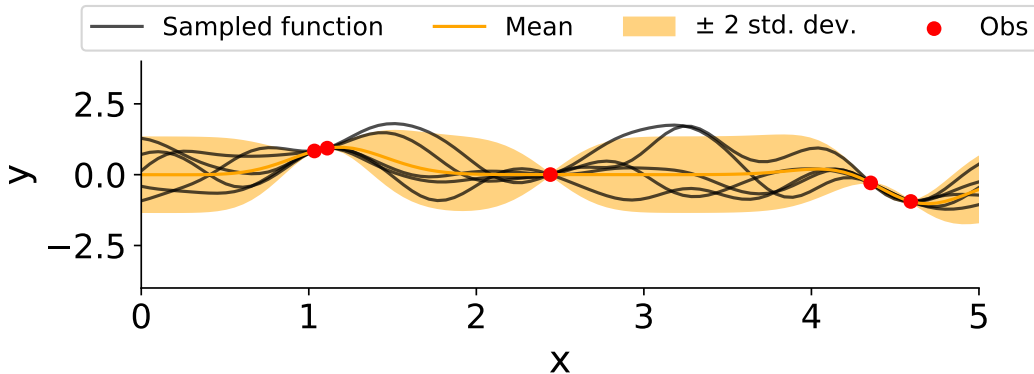
RBF kernel - samples from posterior



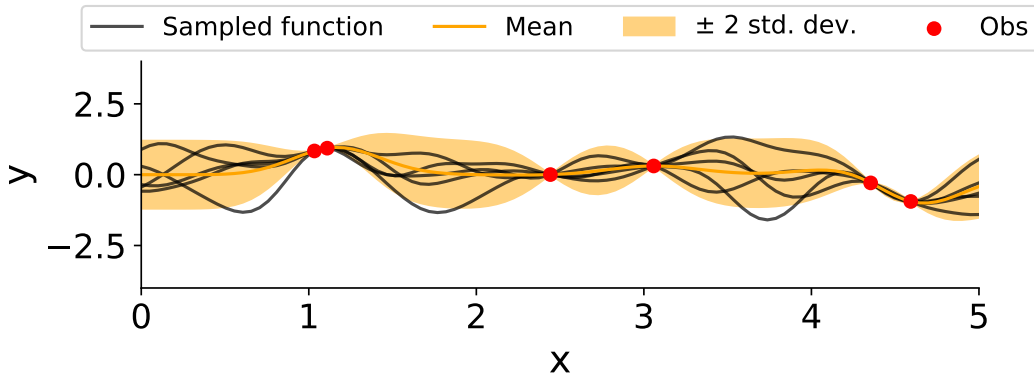
RBF kernel - samples from posterior



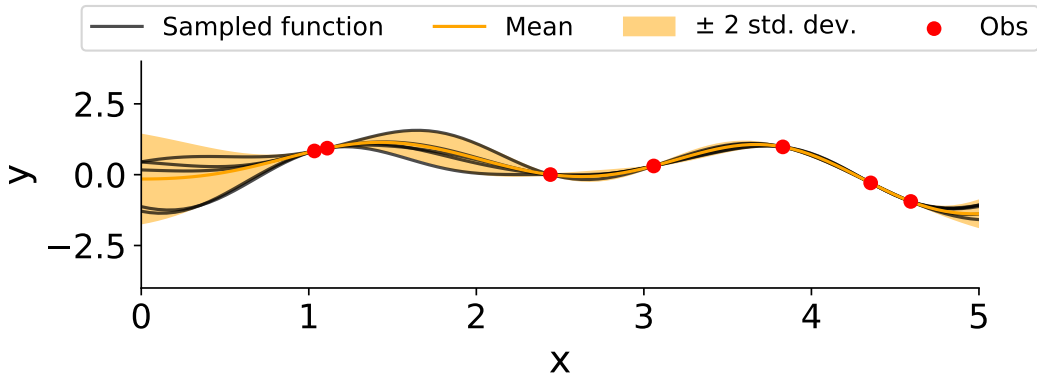
RBF kernel - samples from posterior



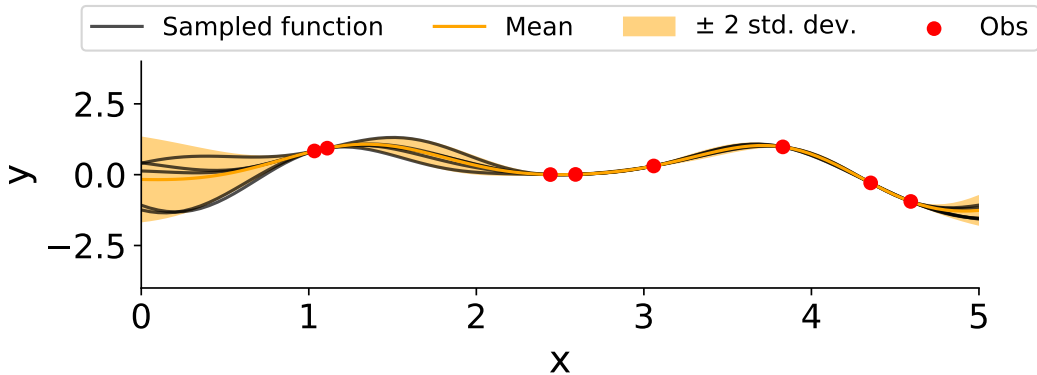
RBF kernel - samples from posterior



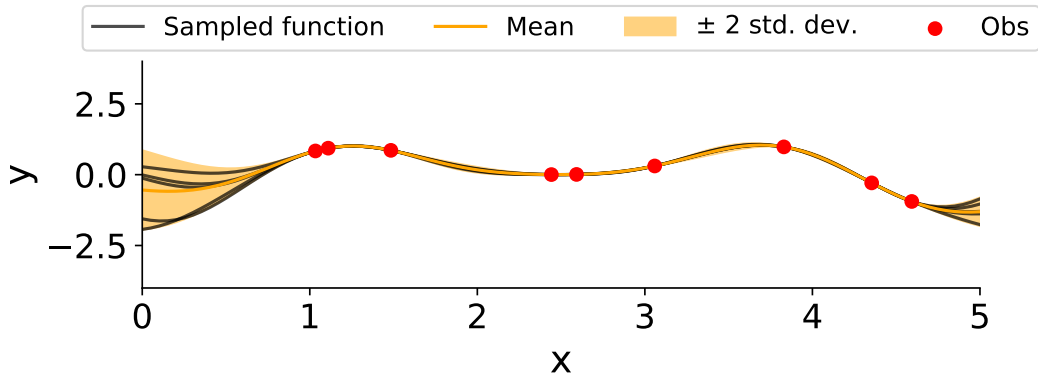
RBF kernel - samples from posterior



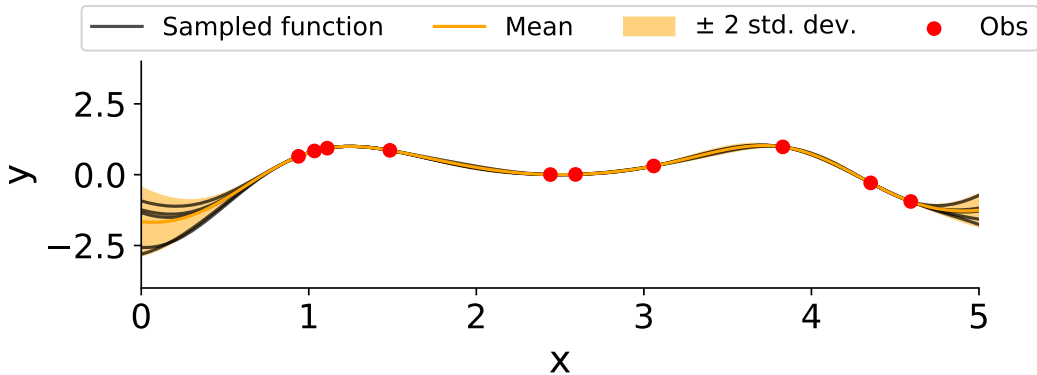
RBF kernel - samples from posterior



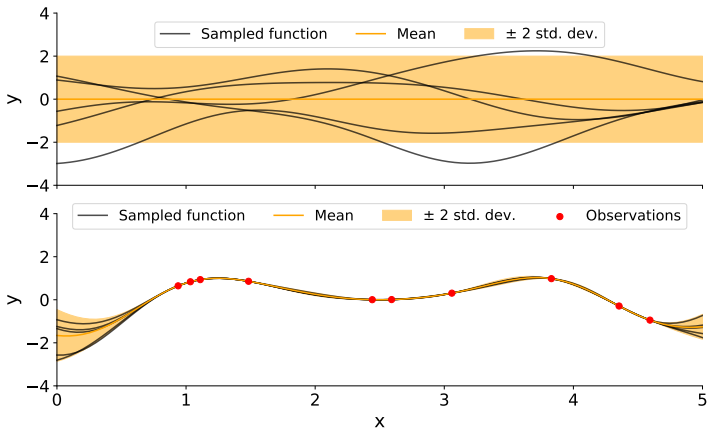
RBF kernel - samples from posterior



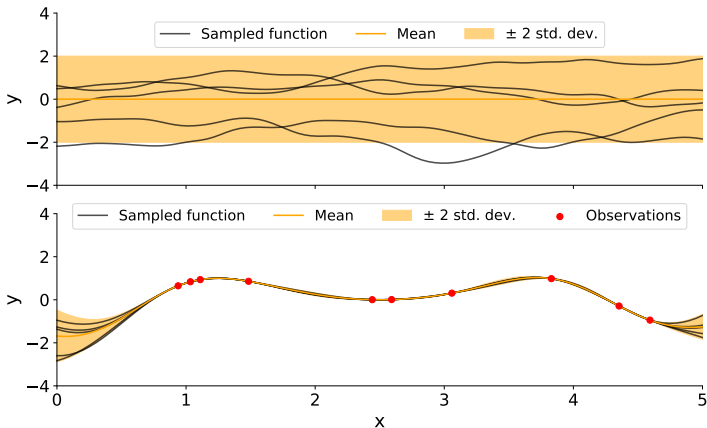
RBF kernel - samples from posterior



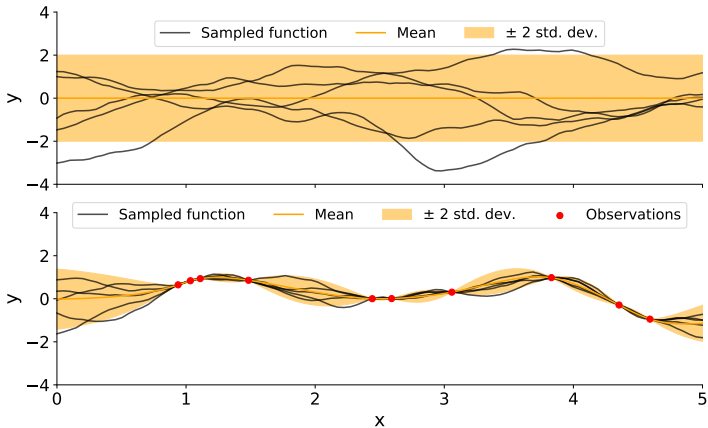
RBF kernel - prior / posterior



Rational-quadratic kernel - prior / posterior



Matern kernel - prior / posterior



3. Stationarity

- **Stationary kernel:** Covariance depends only on the *difference* $\mathbf{x} - \mathbf{x}'$, invariant under translation
- **Non-stationary kernel:** Covariance depends on absolute locations, allowing position-dependent behavior

4. Isotropy

- **Isotropic kernel:** Covariance depends only on the *Euclidean distance* $\|\mathbf{x} - \mathbf{x}'\|$
- **Anisotropic kernel:** Different characteristic length-scales and couplings across dimensions

Stationary Kernels

A stationary covariance function satisfies:

$$\text{Cov}[f(\mathbf{x}), f(\mathbf{x}')] = k(\mathbf{x} - \mathbf{x}') = \psi(\mathbf{x} - \mathbf{x}')$$

Translation invariant: $k(\mathbf{x} + \mathbf{a}, \mathbf{x}' + \mathbf{a}) = k(\mathbf{x}, \mathbf{x}')$ for all shifts $\mathbf{a}$.

Example: RBF kernel

Isotropic Kernels

An isotropic kernel depends only on distance:

$$k(\mathbf{x}, \mathbf{x}') = \phi(\|\mathbf{x} - \mathbf{x}'\|)$$

where $\phi : [0, \infty) \rightarrow \mathbb{R}$ is a scalar function.

Property: Every isotropic kernel is stationary, but not all stationary kernels are isotropic.

Example: RBF kernel is isotropic when $\Lambda = \ell^2 I$ (identity matrix)

Anisotropic Kernels

A stationary *but not isotropic* kernel depends on the full lag vector $\mathbf{h} = \mathbf{x} - \mathbf{x}'$, including directional information:

$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{x}')^\top \Lambda^{-1} (\mathbf{x} - \mathbf{x}') \right)$$

where Λ is a positive definite matrix defining different characteristic length-scales and couplings across dimensions.

Use case: When different input dimensions have different “importance” or correlation decay rates.

Non-Stationary Kernels

Covariance depends on absolute locations, not just their difference. Allow position-dependent behavior and varying smoothness.

Examples: Polynomial kernel, neural network kernel

Kernel: Squared-Exponential (RBF)

1D Form:

$$k(x, x') = \sigma_f^2 \exp \left(-\frac{(x - x')^2}{2\ell^2} \right)$$

Multi-dimensional isotropic form:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp \left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2} \right)$$

Characteristics: Infinitely differentiable (very smooth sample functions), **stationary and isotropic**, rapid decay with distance, widely used due to simplicity and smoothness.

Parameters: σ_f^2 , signal variance (amplitude); ℓ , length-scale.

Kernel: Matérn

General form:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{\|\mathbf{x} - \mathbf{x}'\|}{\ell} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{\|\mathbf{x} - \mathbf{x}'\|}{\ell} \right)$$

where K_ν is a modified Bessel function, $\nu > 0$ controls smoothness, and ℓ is the length-scale.

Common versions: $\nu = 1/2$, exponential kernel; $\nu = 3/2$; $\nu \rightarrow \infty$, approaches RBF kernel.

Characteristics: More flexible smoothness control than RBF, includes exponential kernel as special case, stationary and isotropic.

Parameters: σ_f^2 , signal variance; ℓ , length-scale; ν , smoothness parameter.

Kernel: Rational Quadratic (RQ)

Form:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \left(1 + \frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\alpha\ell^2} \right)^{-\alpha}$$

Characteristics: Mixture of RBF kernels with different length-scales; can be viewed as weighted superposition of exponential kernels at different scales, **stationary and isotropic**.

Parameters: σ_f^2 , signal variance, ℓ , length-scale; α , mixing parameter (larger α , closer to RBF).

Kernel: Periodic

Form:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp \left(-\frac{2 \sin^2(\pi |\mathbf{x} - \mathbf{x}'| / p)}{\ell^2} \right)$$

Characteristics: Captures periodic patterns in data; essential for time-series with seasonal patterns.

Parameters: σ_f^2 : signal variance; ℓ : length-scale; p : period (wavelength).

Kernel: Polynomial

Form:

$$k(\mathbf{x}, \mathbf{x}') = \left(\sigma_f^2 \mathbf{x} \cdot \mathbf{x}' + c \right)^d$$

or

$$k(\mathbf{x}, \mathbf{x}') = \left(\sigma_f^2 \mathbf{x} \cdot \mathbf{x}' + c \right)^d + \sigma_n^2$$

Characteristics: Often used with small d (typically 1, 2, or 3),
non-stationary kernel.

Parameters: σ_f^2 , signal variance, c , offset, d , degree (model complexity), σ_n^2 , noise variance (if included).

Kernel: White Noise

Form:

$$k(x, x') = \sigma_n^2 \delta(x - x')$$

where $\delta(\cdot)$ is the Dirac delta function.

In practice (discrete inputs with finite precision):

$$k_{\text{white noise}}(\mathbf{x}, \mathbf{x}') = \begin{cases} \sigma_n^2 & \text{if } \mathbf{x} = \mathbf{x}' \\ 0 & \text{otherwise} \end{cases}$$

Kernel: White Noise

- **Pure noise:** No correlation between any two distinct points.
- **Variance only on diagonal:** Contributes σ_n^2 to the diagonal of the covariance matrix.
- **Numerical stability:** Often added to smooth kernels to stabilize matrix inversion.
- **Independent observations:** Assumes each observation contains independent measurement noise.

Kernel: White Noise

1. **Likelihood noise term:** Observations $y_i = f(\mathbf{x}_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$
2. **Jitter for numerical stability:** Added to other kernels to improve matrix conditioning:

$$k_{\text{total}} = k_{\text{smooth}}(\mathbf{x}, \mathbf{x}') + \sigma_n^2 \delta(\mathbf{x} - \mathbf{x}')$$

3. **Model white noise in data:** When you expect uncorrelated noise in observations

Kernel combinations

Kernels are closed under addition and multiplication:

- **Sum (additive):** $k_{\text{total}} = k_1 + k_2$ (combines features)
 - Example: RBF + Periodic (smooth + periodic patterns)
 - Example: RBF + White Noise (smooth signal + independent noise)
- **Product (multiplicative):** $k_{\text{total}} = k_1 \times k_2$ (modulates one kernel by another)
 - Example: (RBF) $\times$ (Periodic) (smooth periodic trends)

Estimating the kernel parameters

$$K_y(x_p, x_q) = \sigma_f^2 \exp\left(-\frac{1}{2\ell^2} (x_p - x_q)^2\right) + \sigma_y^2 \delta_{pq}$$

We can maximize the marginal likelihood:

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{f}, \mathbf{X}) p(\mathbf{f} \mid \mathbf{X}) d\mathbf{f}$$

Estimating the kernel parameters

We can maximize the marginal likelihood:

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{f}, \mathbf{X}) p(\mathbf{f} \mid \mathbf{X}) d\mathbf{f}$$

where:

$$p(\mathbf{f} \mid \mathbf{X}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K})$$

$$p(\mathbf{y} \mid \mathbf{f}) = \prod_i \mathcal{N}(y_i \mid f_i, \sigma_y^2)$$

Estimating the kernel parameters

The marginal likelihood is given by:

$$\begin{aligned}\log p(\mathbf{y} \mid \mathbf{X}) &= \log \mathcal{N}(\mathbf{y} \mid \mathbf{0}, \mathbf{K}_y) \\ &= - \underbrace{\frac{1}{2} \mathbf{y} \mathbf{K}_y^{-1} \mathbf{y}}_{\text{data fit}} - \underbrace{\frac{1}{2} \log |\mathbf{K}_y|}_{\text{model complexity}} - \underbrace{\frac{N}{2} \log(2\pi)}_{\text{constant}}\end{aligned}$$

Estimating the kernel parameters

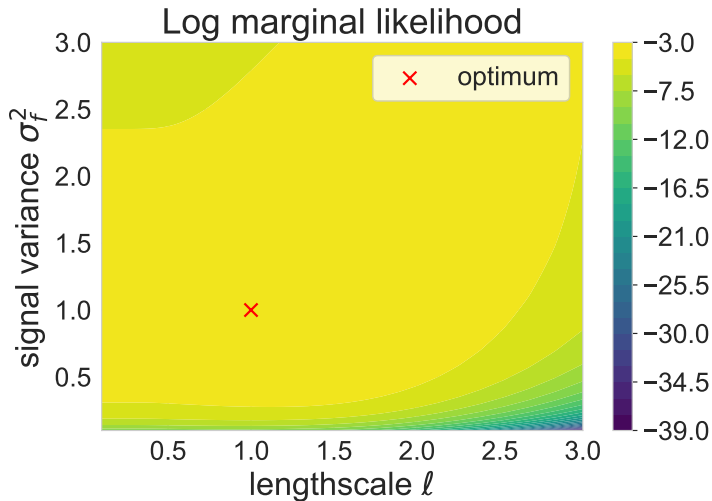
Let the kernel parameters (also called hyper-parameters) be denoted by θ .

One can show that:

$$\begin{aligned}\frac{\partial}{\partial \theta_j} \log p(\mathbf{y} \mid \mathbf{X}) &= \frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \frac{\partial \mathbf{K}_y}{\partial \theta_j} \mathbf{K}_y^{-1} \mathbf{y} - \frac{1}{2} \text{tr} \left(\mathbf{K}_y^{-1} \frac{\partial \mathbf{K}_y}{\partial \theta_j} \right) \\ &= \frac{1}{2} \text{tr} \left((\boldsymbol{\alpha} \boldsymbol{\alpha}^T - \mathbf{K}_y^{-1}) \frac{\partial \mathbf{K}_y}{\partial \theta_j} \right)\end{aligned}$$

where $\boldsymbol{\alpha} = \mathbf{K}_y^{-1} \mathbf{y}$.

Log marginal likelihood



Computational cost

- One difficulty with GPs is the computational cost of training them: $O(n^3)$ (and $O(n^2)$ memory).
- They work out of the box for n in the order of a few thousands.
- There are many ways to side-step this cost: inducing inputs, efficient matrix-vector multiplications, random features, etc.
- These days we can use GPs for n in the order of tens of millions.

Applications

Automatic Relevance Determination (ARD)

Multi-task Gaussian processes

Gaussian processes and neural networks

Deconvolution using Gaussian processes

Automatic Relevance Determination (ARD)

- **Automatic Relevance Determination (ARD)** is a technique for determining the importance of input variables.
- Originally formulated in **neural networks** (MacKay 1994, Neal 1996).
- Extensively used in **Gaussian Process** regression.
- Optimize separate length-scale parameters for each input dimension.

ARD Kernel Function

Standard squared-exponential kernel with ARD:

$$k(x_n, x_m) = \sigma_0 \exp \left(-\frac{1}{2} \sum_{i=1}^D \lambda_i (x_{ni} - x_{mi})^2 \right) + \sigma_3 x_n^T x_m$$

- σ_0 : overall variance
- λ_i : **precision for dimension i**
- σ_3 : linear term
- If λ_i is **large**, dimension i is **important**
- If λ_i is **small**, dimension i is **irrelevant**

ARD Kernel Function

Large λ_i (Relevant)

Steep kernel in dimension i

Function changes rapidly with x_i

Strong dependence on input i

Small λ_i (Irrelevant)

Flat kernel in dimension i

Function changes slowly with x_i

Weak/no dependence on input i

Example: $\lambda_1 = 1$ vs $\lambda_2 = 0.01$

- Function is sensitive to x_1 .
- Function is nearly insensitive to x_2 .

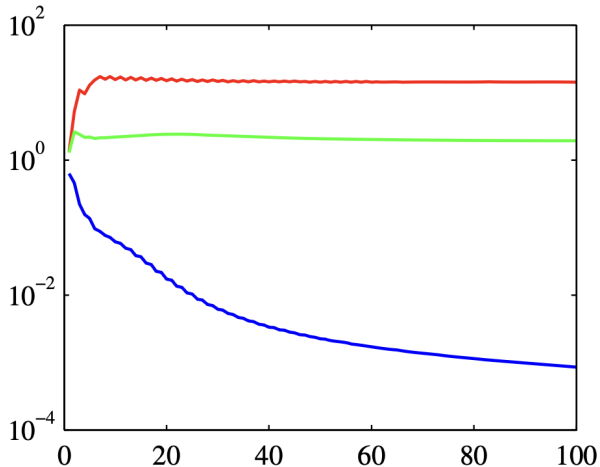
ARD

Synthetic Dataset:

- x_1 : generated from Gaussian.
- **Target:** $t = \sin(2x_1) + \text{Gaussian noise}$.
- x_2 : copy of $x_1 + \text{noise}$ (partially correlated).
- x_3 : independent Gaussian (irrelevant).

Example Results

λ_1 (red), λ_2 (green), λ_3 (blue)



General Form with Linear Terms

Extended ARD kernel (for regression):

$$k(x_n, x_m) = \sigma_0 \exp \left(-\frac{1}{2} \sum_{i=1}^D \lambda_i (x_{ni} - x_{mi})^2 \right) + \sigma_3 \sum_{i=1}^D x_{ni} x_{mi}$$

Two competing mechanisms:

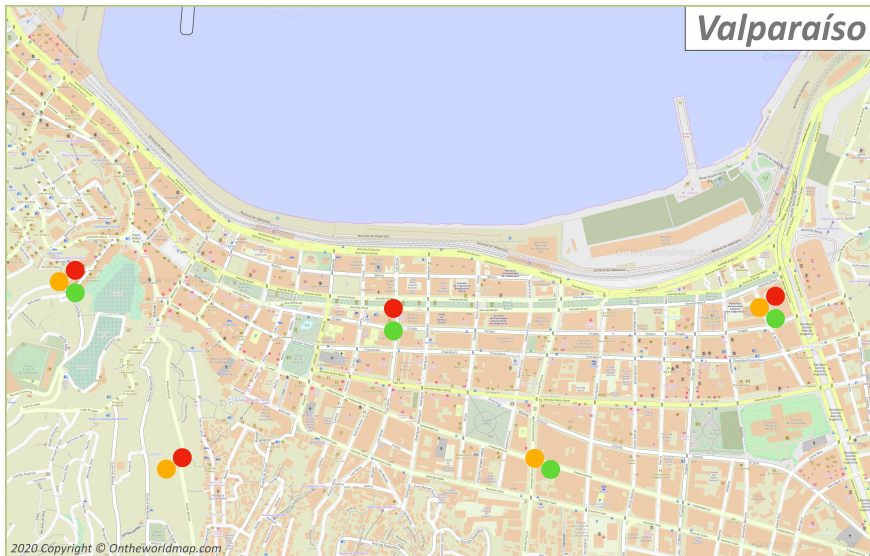
1. **Exponential term** with separate λ_i parameters
2. **Linear term** with fixed coupling

Connection to other methods

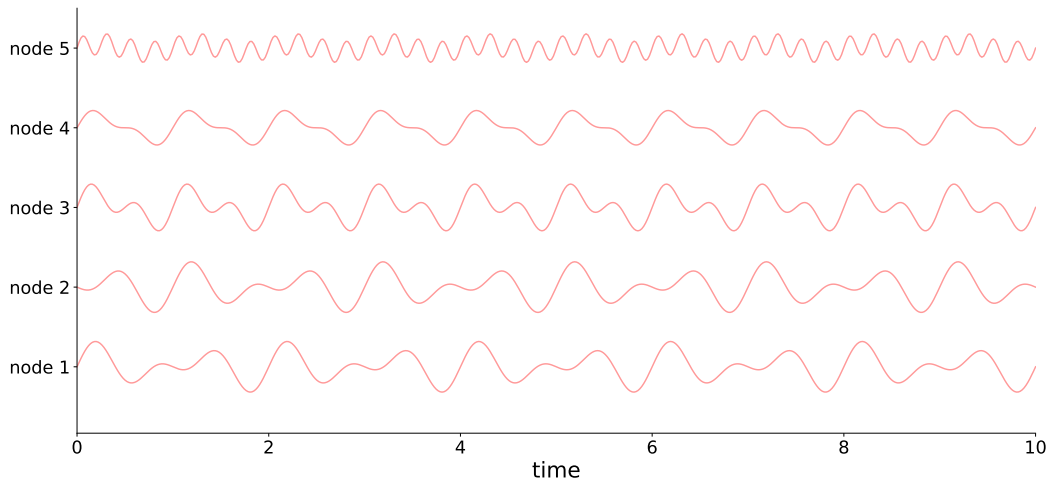
Method	Approach
Feature Selection	Discrete: keep or remove feature
Regularization (L1)	Shrink weights toward zero
ARD in Bayesian NNs	Prune weights/hyperparameters
ARD in GPs	Automatically scale input dimensions

Multi-task or multi-output GPs

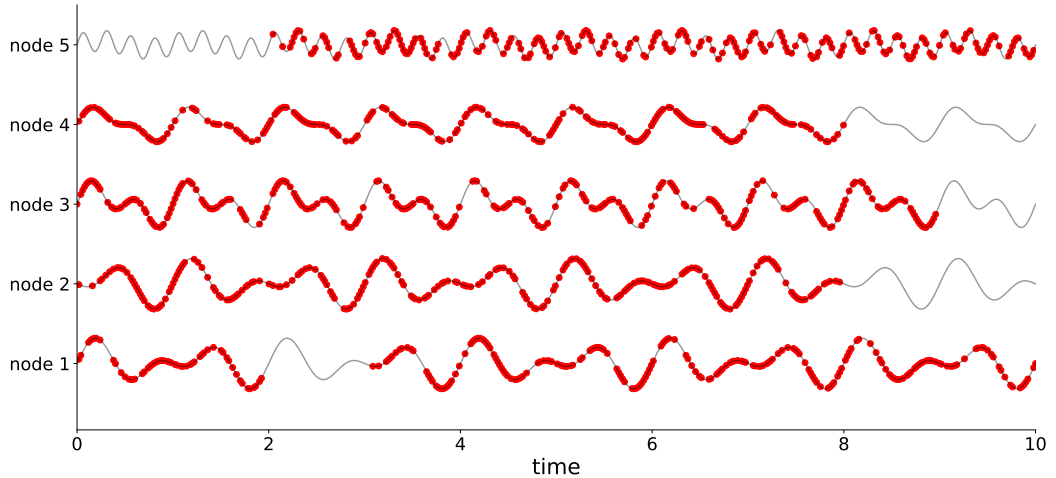
Multi-task GPs



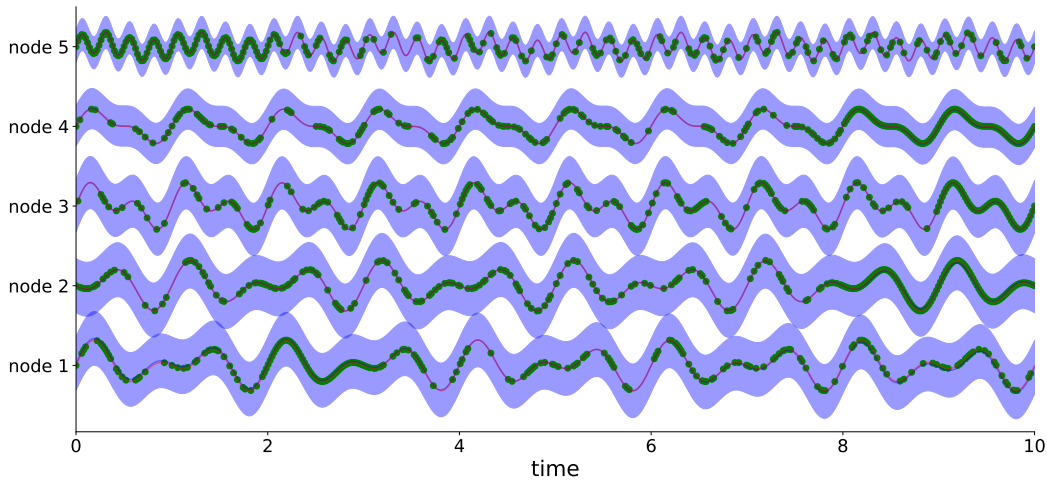
Sensors network



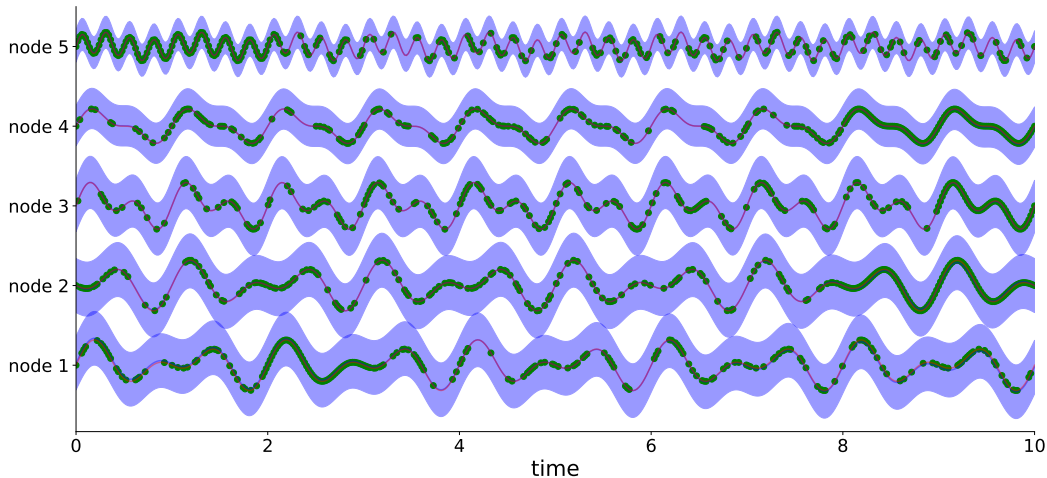
Sensors network with missing segments



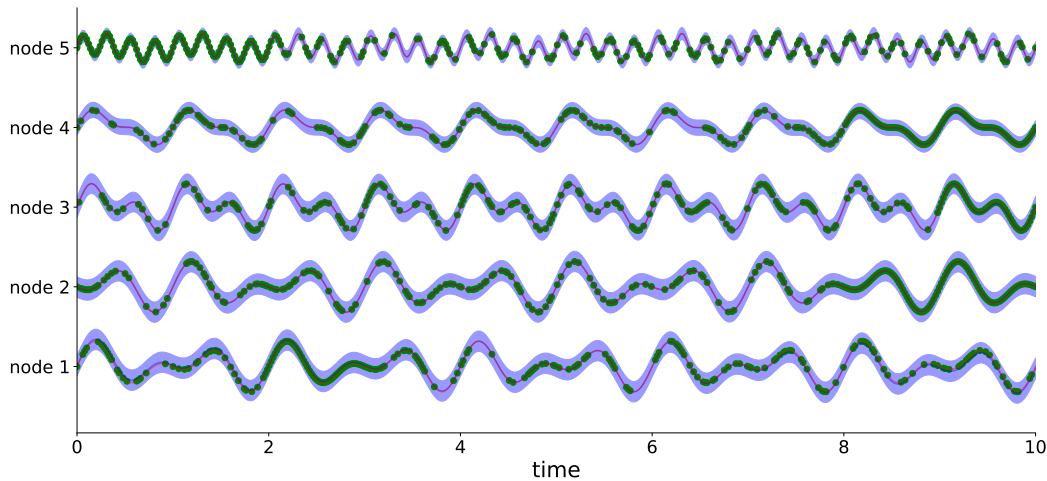
Sensors network - prediction



Sensors network - prediction before training

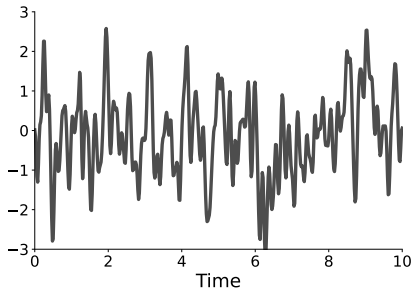


Sensors network - prediction after training



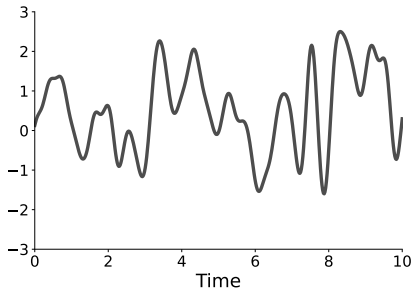
Single-output Gaussian process

$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

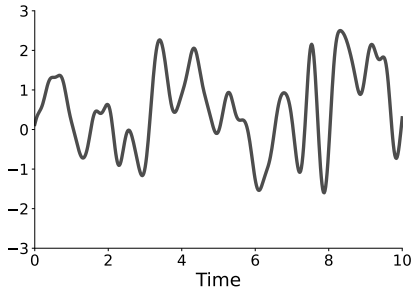


Single-output Gaussian process

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Single-output Gaussian process



$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

Training dataset:

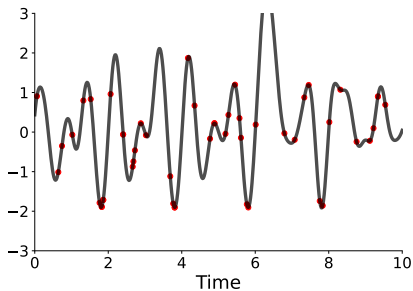
$$\mathcal{D} = \{(\mathbf{x}_i, f(\mathbf{x}_i)) \mid i = 1, \dots, N\}$$

Single-output Gaussian process

$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

Training dataset:

$$\mathcal{D} = \{(\mathbf{x}_i, f(\mathbf{x}_i)) \mid i = 1, \dots, N\}$$



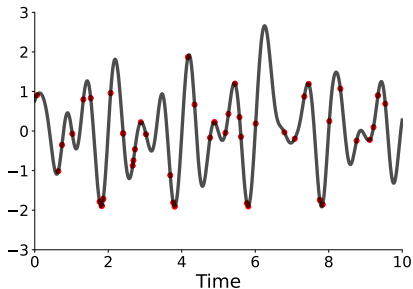
$$\begin{bmatrix} f(\mathbf{x}_1) \\ \vdots \\ f(\mathbf{x}_N) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

Single-output Gaussian process

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$$\begin{bmatrix} f(\mathbf{x}_1) \\ \vdots \\ f(\mathbf{x}_N) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

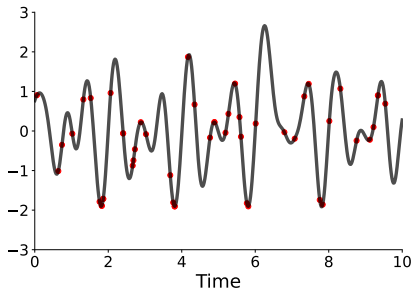
In matrix form: $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$

Single-output Gaussian process

$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

Training dataset:

$$\mathcal{D} = \{(\mathbf{x}_i, f(\mathbf{x}_i)) \mid i = 1, \dots, N\}$$

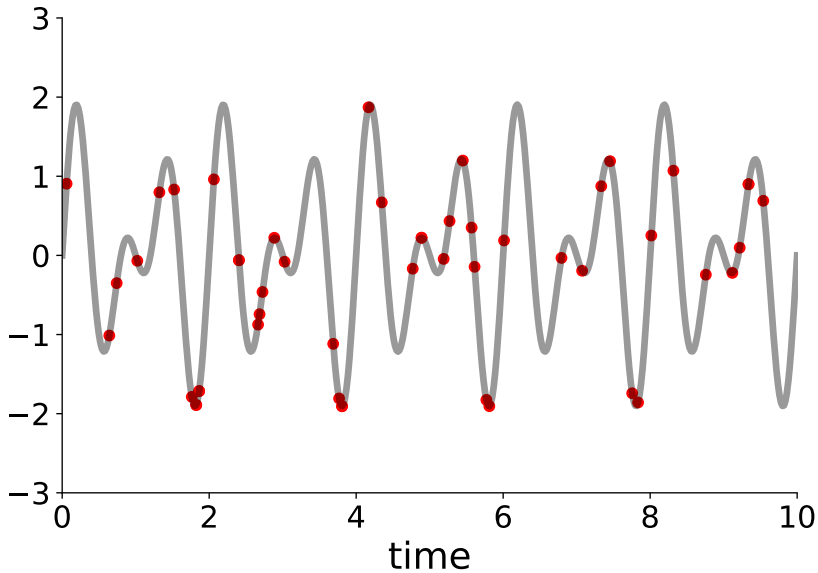


$$\begin{bmatrix} f(\mathbf{x}_1) \\ \vdots \\ f(\mathbf{x}_N) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

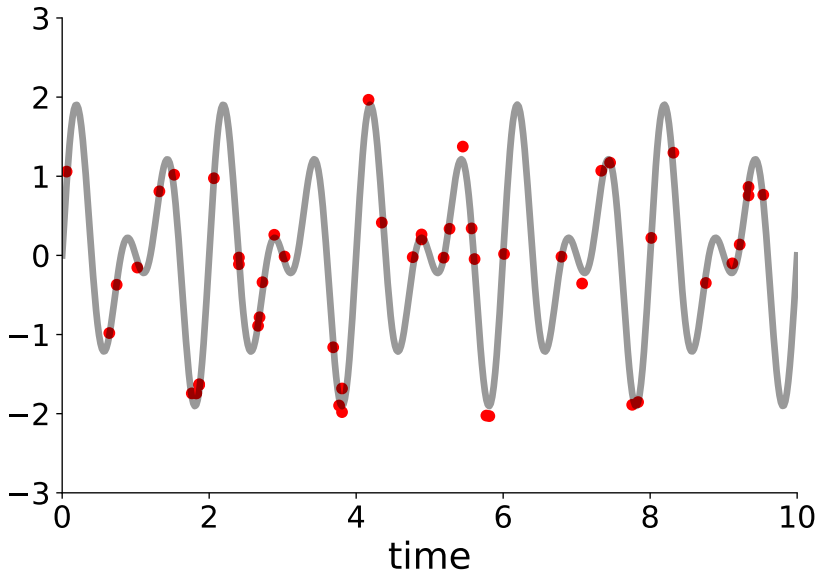
In matrix form: $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$

For prediction: $p(f(\mathbf{x}_*) \mid \mathbf{f})$

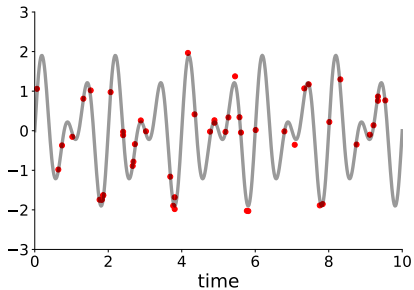
Noisy observations



Noisy observations

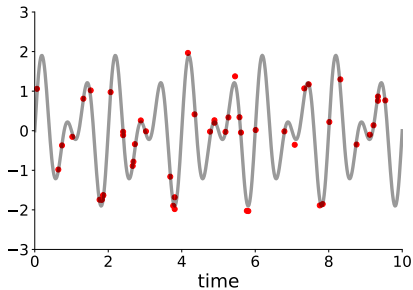


GP with noisy observations



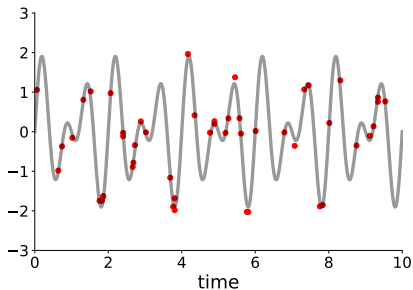
$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

GP with noisy observations



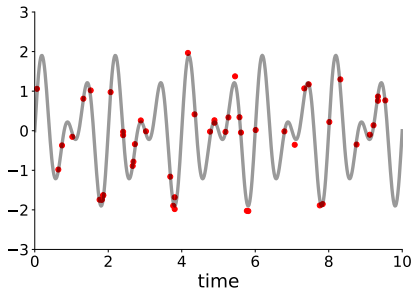
$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$
$$y(\mathbf{x}_i) = f(\mathbf{x}_i) + \epsilon_i$$

GP with noisy observations



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$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

GP with noisy observations



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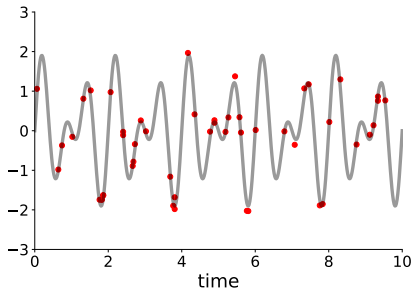
$$y(\mathbf{x}_i) = f(\mathbf{x}_i) + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

$$\mathcal{D} = \{(\mathbf{x}_i, y(\mathbf{x}_i)) \mid i = 1, \dots, N\}$$

$$\begin{bmatrix} y(\mathbf{x}_1) \\ \vdots \\ y(\mathbf{x}_N) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} + \sigma^2 \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right)$$

GP with noisy observations



$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

$$y(\mathbf{x}_i) = f(\mathbf{x}_i) + \epsilon_i$$

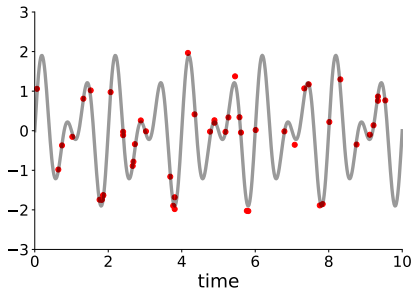
$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

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In matrix form: $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K} + \sigma^2 \mathbf{I})$

GP with noisy observations



$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

$$y(\mathbf{x}_i) = f(\mathbf{x}_i) + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

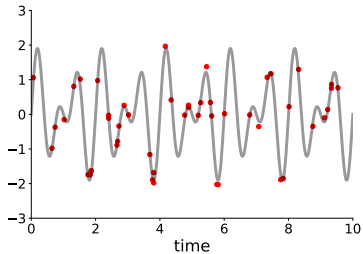
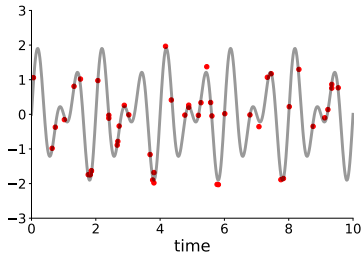
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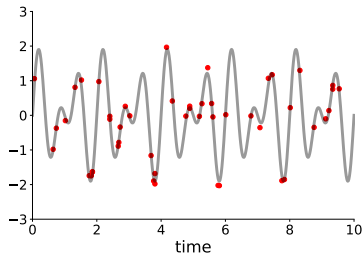
In matrix form: $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K} + \sigma\mathbf{I})$

For prediction: $p(f(\mathbf{x}_*) \mid \mathbf{y})$

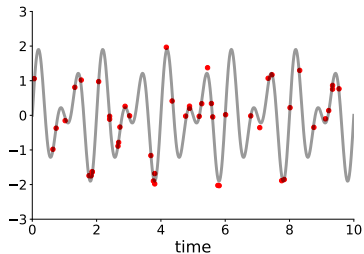
Multiple-output Gaussian process



Multiple-output Gaussian process

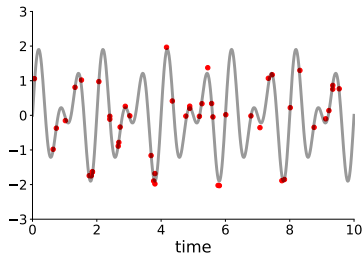


$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$



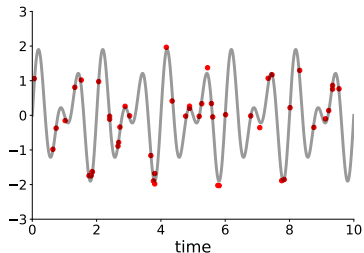
$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$

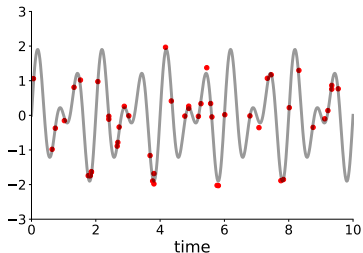
$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, f_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, f_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

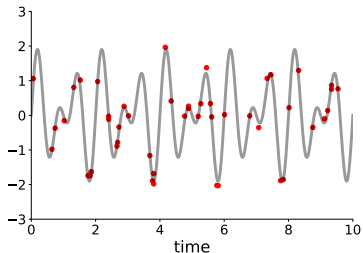
Multiple-output Gaussian process



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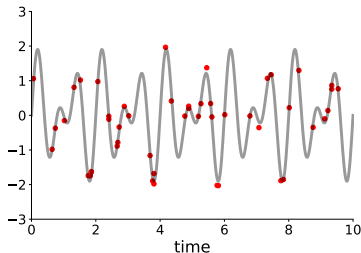


$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

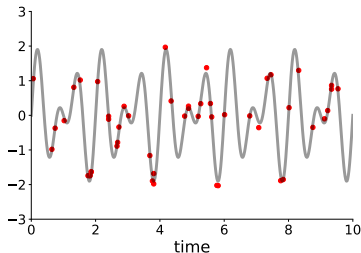
$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, f_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

$$\mathbf{f}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_2)$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, f_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$

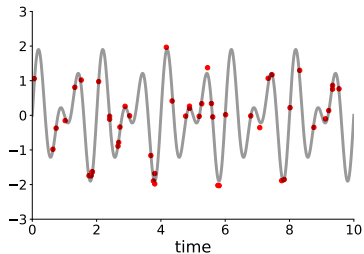
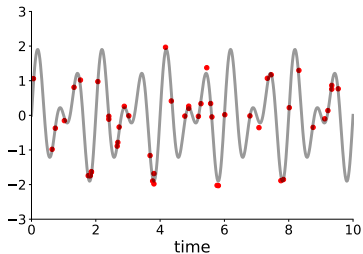
$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, f_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

$$\mathbf{f}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_1)$$

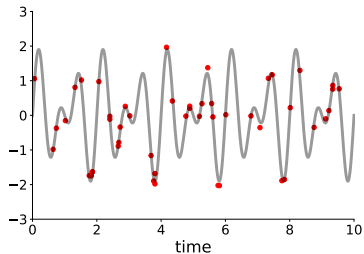
$$\mathbf{f}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_2)$$

$$\underbrace{\begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{bmatrix}}_{\mathbf{f}} \sim \mathcal{N}\left(\underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{0}}, \underbrace{\begin{bmatrix} \mathbf{K}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 \end{bmatrix}}_{\mathbf{K}_{\mathbf{f},\mathbf{f}}}\right)$$

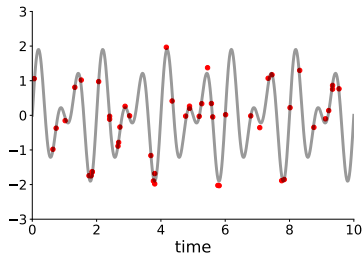
Multiple-output Gaussian process



Multiple-output Gaussian process

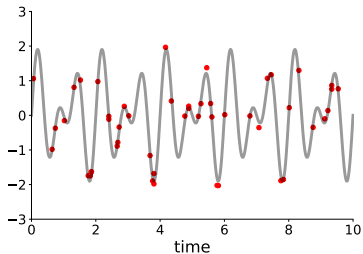


$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$



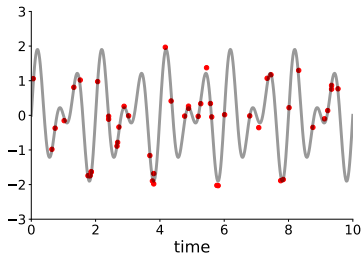
$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$

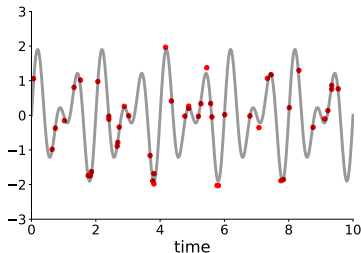
$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, \mathbf{y}_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, \mathbf{y}_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

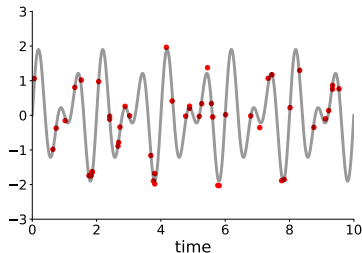
Multiple-output Gaussian process



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$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, \mathbf{y}_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$

$$\mathbf{y}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_1 + \sigma_1^2 \mathbf{I})$$

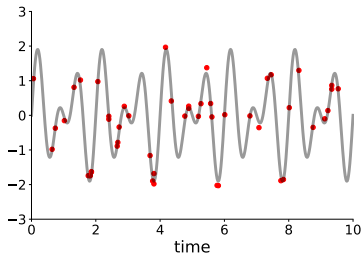


$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

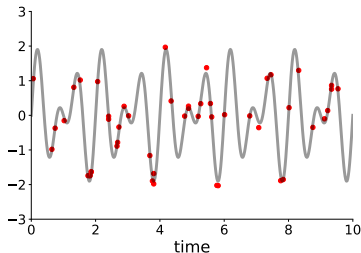
$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, \mathbf{y}_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

$$\mathbf{f}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_2 + \sigma_2^2 \mathbf{I})$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, \mathbf{y}_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$

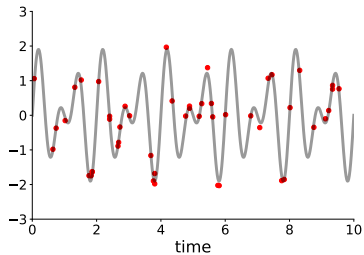
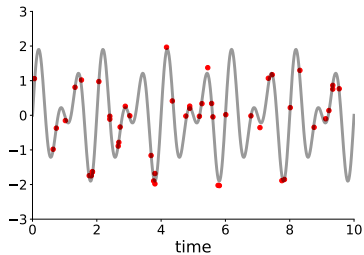
$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, \mathbf{y}_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

$$\mathbf{y}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_1 + \sigma_1^2 \mathbf{I})$$

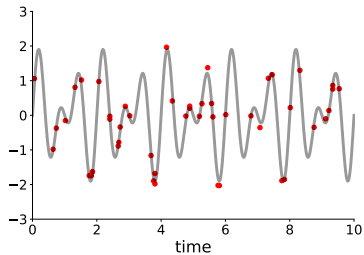
$$\mathbf{f}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_2 + \sigma_2^2 \mathbf{I})$$

$$\underbrace{\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix}}_{\mathbf{y}} \sim \mathcal{N} \left(\underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{0}}, \underbrace{\begin{bmatrix} \mathbf{K}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 \end{bmatrix}}_{\mathbf{K}_{f,f}} + \underbrace{\begin{bmatrix} \sigma_1^2 \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \sigma_2^2 \mathbf{I} \end{bmatrix}}_{\Sigma} \right)$$

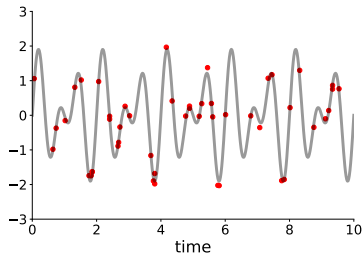
Multiple-output Gaussian process



Multiple-output Gaussian process

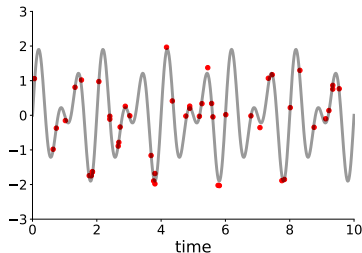


$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$



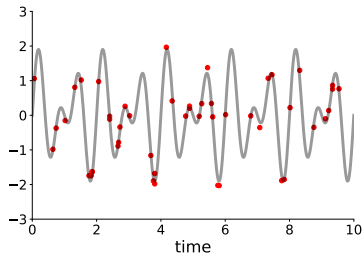
$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$

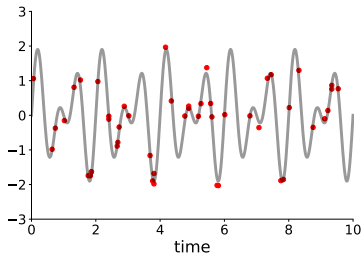
$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, f_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

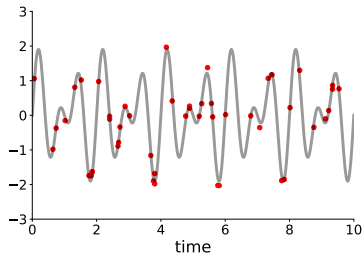
$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, f_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

Multiple-output Gaussian process



$$f_1(\mathbf{x}) \sim \mathcal{GP}(0, k_1(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_1 = \{(\mathbf{x}_{i,1}, f_1(\mathbf{x}_{i,1})) \mid i = 1, \dots, N_1\}$$



$$f_2(\mathbf{x}) \sim \mathcal{GP}(0, k_2(\mathbf{x}, \mathbf{x}'))$$

$$\mathcal{D}_2 = \{(\mathbf{x}_{i,2}, f_2(\mathbf{x}_{i,2})) \mid i = 1, \dots, N_2\}$$

$$\mathbf{K}_{f,f} = \begin{bmatrix} \mathbf{K}_1 & ? \\ ? & \mathbf{K}_2 \end{bmatrix}$$

Build a cross-covariance function

$\text{cov}[f_1(\mathbf{x}), f_2(\mathbf{x}')] such that $\mathbf{K}_{f,f}$ is positive semi-definite.$

Why model task correlations?

Many applications: multi-output sensors, multi-channel time series, multi-label prediction.

If tasks are correlated, sharing structure can:

- Improve predictions for low-data tasks.
- Regularize hyperparameters and reduce overfitting.
- Provide coherent multi-output uncertainty.

Multi-task GPs

Multi-task Gaussian Processes (GPs) model multiple correlated outputs jointly.

Models:

- Intrinsic Coregionalization Model (ICM).
- Linear Model of Coregionalization (LMC).

The idea is to exploit task correlations to improve prediction and uncertainty quantification.

Vector-valued GP

Multi-task GP: $f : \mathcal{X} \rightarrow \mathbb{R}^D$, $f(x) = (f_1(x), \dots, f_D(x))^T$.

Mean function:

$$m : \mathcal{X} \rightarrow \mathbb{R}^D.$$

Matrix-valued kernel:

$$k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^{D \times D},$$

encoding both input similarity and inter-task covariance.

Independent multi-task baseline

Simplest approach: one independent GP per task

$i = 1, \dots, D$:

$$f_i(X) \sim \mathcal{N}(m_i(X), K_i(X, X)).$$

Joint covariance (stacking all tasks) is block-diagonal:

$$K(X, X) = \text{diag}(K_1(X, X), \dots, K_D(X, X)).$$

No information sharing across tasks; correlations are ignored.

Intrinsic Coregionalization Model

Consider D outputs collected in $f(x) = (f_1(x), \dots, f_D(x))^T$.

ICM assumes:

$$f(x) \sim \mathcal{GP}(0, K((x, i), (x', j))),$$

with separable covariance

$$\text{Cov}(f_i(x), f_j(x')) = B_{ij} k(x, x').$$

where $k(x, x')$ is a scalar positive definite kernel, and $B \in \mathbb{R}^{D \times D}$ is symmetric positive semidefinite.

ICM kernel as Kronecker product

Let $X = (x_1, \dots, x_N)$ and $K_X \in \mathbb{R}^{N \times N}$ with

$$(K_X)_{nm} = k(x_n, x_m).$$

Stack all outputs as $\text{vec}(F) \in \mathbb{R}^{ND}$, where $F_{n,i} = f_i(x_n)$.

Under ICM:

$$\text{vec}(F) \sim \mathcal{N}(0, K_{\text{ICM}}), \quad K_{\text{ICM}} = B \otimes K_X.$$

This shows a separable structure: task covariance $\times$ input covariance.

ICM: positive semidefiniteness of B

B must be positive semidefinite: $z^\top B z \geq 0$ for all $z \in \mathbb{R}^D$.

A common parameterization:

$$B = WW^\top + \text{diag}(v),$$

with $W \in \mathbb{R}^{D \times R}$ (low rank) and $v \in \mathbb{R}_{\geq 0}^D$.

This guarantees $B \succeq 0$ and controls rank of inter-task correlations via R and marginal variances via v .

ICM: correlation coefficients

Covariance between tasks i, j at a fixed input:

$$\text{Cov}(f_i(x), f_j(x)) = B_{ij} k(x, x).$$

If $k(x, x) = \sigma_k^2$, the task covariance (up to σ_k^2) is B_{ij} .

Pearson correlation coefficient:

$$\rho_{ij} = \frac{B_{ij}}{\sqrt{B_{ii} B_{jj}}}.$$

ICM: latent-factor construction (rank-1)

Single latent GP:

$$u(x) \sim \mathcal{GP}(0, k(x, x')).$$

Linear mixing to outputs:

$$f_i(x) = a_i u(x), \quad i = 1, \dots, D.$$

Then

$$\text{Cov}(f_i(x), f_j(x')) = a_i a_j k(x, x'),$$

Hence $B = aa^\top$, which is rank-1 and positive semidefinite.

ICM: low-rank latent-factor view

Generalize to R latent GPs $u_r(x) \sim \mathcal{GP}(0, k(x, x'))$, independent over r .

Mixing:

$$f_i(x) = \sum_{r=1}^R w_{ir} u_r(x),$$

with $W = (w_{ir}) \in \mathbb{R}^{D \times R}$.

Then

$$\text{Cov}(f_i(x), f_j(x')) = \sum_{r=1}^R w_{ir} w_{jr} k(x, x') = B_{ij} k(x, x'),$$

with $B = WW^\top$.

LMC: generalization

LMC allows multiple input kernels with latent GPs

$u_q(x) \sim \mathcal{GP}(0, k_q(x, x'))$, independent over $q = 1, \dots, Q$.

Mixing:

$$f_i(x) = \sum_{q=1}^Q a_{iq} u_q(x).$$

Then

$$\text{Cov}(f_i(x), f_j(x')) = \sum_{q=1}^Q a_{iq} a_{jq} k_q(x, x') = \sum_{q=1}^Q B_{ij}^{(q)} k_q(x, x'),$$

with $B^{(q)} = a_{:,q} a_{:,q}^\top$.

ICM as special case of LMC

If all latent processes share the same kernel k , i.e. $k_q = k$ for all q :

$$\text{Cov}(f_i(x), f_j(x')) = \left(\sum_{q=1}^Q B_{ij}^{(q)} \right) k(x, x') = B_{ij} k(x, x'),$$

where $B = \sum_{q=1}^Q B^{(q)}$.

- ICM typically refers to a single shared kernel and one coregionalization matrix, while LMC uses sums of such components.
- LMC can capture richer patterns when tasks behave differently across inputs.

Matrix form and structure

Under ICM, for N inputs and D tasks:

$$\text{vec}(F) \sim \mathcal{N}(0, B \otimes K_X).$$

Outputs are coupled via B and share smoothness through k .

Under LMC:

$$K_{\text{LMC}} = \sum_{q=1}^Q B^{(q)} \otimes K_X^{(q)},$$

with $(K_X^{(q)})_{nm} = k_q(x_n, x_m)$.

Noise modeling

Observations:

$$y(x) = f(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}_D(0, \Sigma),$$

with $\Sigma \in \mathbb{R}^{D \times D}$ positive semidefinite.

Choices: - Diagonal: $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_D^2)$. - Full: $\Sigma = LL^\top$
with $L \in \mathbb{R}^{D \times r}$.

Correlated noise models shared disturbances across tasks.

Joint covariance with noise

Let K be the noise-free covariance from ICM/LMC over all tasks and inputs.

For N inputs and D tasks (stacked), a common form is:

$$K_n = K + I_N \otimes \Sigma,$$

where I_N is the $N \times N$ identity matrix.

The Kronecker structure is preserved when Σ is diagonal; with full Σ structure is more complex but still interpretable.

Predictions and uncertainty

At test inputs X_* :

$$p(f(X_*) \mid Y) = \mathcal{N}(\mu_*, \Sigma_*).$$

For each task t :

- Predictive mean $\mu_{*,t}(x)$.
- Variance $\text{Var}(f_t(x))$ for credible bands.

Joint Σ_* contains cross-task predictive covariances.

Learned B and Σ

Coregionalization matrix:

$$B = WW^T + \text{diag}(v),$$

learned from data.

Interpretation:

- B_{ii} is the marginal variance of task i .
- B_{ij} is the learned covariance between tasks i and j .

Noise covariance $\Sigma = LL^T$ can also be inspected (e.g. heatmaps) to understand shared noise.

Independent GPs vs ICM

Independent GPs:

- Separate mean, kernel, and likelihood per task.
- No cross-task covariance; hyperparameters estimated from each task alone.

ICM:

- Shared input kernel hyperparameters.
- Cross-task structure encoded in B and possibly Σ .

With few observations per task, ICM can regularize and improve generalization.

When to use ICM vs LMC

ICM (single shared kernel):

- Tasks have similar smoothness and variation across input space.
- Fewer hyperparameters, simpler optimization.

LMC (multiple kernels):

- Tasks differ in smoothness, periodicity, or characteristic scales.
- More flexible but higher computational and optimization cost.

Practical strategy: Start with low-rank ICM; increase rank or move to LMC if cross-task structure appears underfitted.

Infinite Neural Network = Gaussian Process

Single Hidden Layer Architecture

One-Hidden-Layer Network

Network structure:

- Input: $\mathbf{x} \in \mathbb{R}^d$
- Hidden layer: J units producing $\mathbf{a}^{(1)} \in \mathbb{R}^J$
- Output layer: single linear unit producing $y_k \in \mathbb{R}$

Forward pass:

$$y_k = f(\mathbf{x}; \mathbf{w}) = \sum_{j=1}^J w_{kj} h \left(\sum_i w_{ji} x_i + w_{j0} \right) + w_{k0}$$

where:

- $h(\cdot)$ is the activation function (e.g., ReLU, tanh).
- w_{ji}, w_{j0} : weights and bias of hidden unit j .
- w_{kj}, w_{k0} : weights and bias of the single output neuron.

Data Flow: Inputs and Outputs

Component	Input(s)	Output
Single neuron	$\mathbf{x}$	scalar activation a
Single hidden layer	$\mathbf{x}$	vector $\mathbf{a}^{(1)} \in \mathbb{R}^J$
Whole finite NN	$\mathbf{x}$	scalar prediction $y_k = f(\mathbf{x}; \mathbf{w})$

Training

Weights $\mathbf{w}$, $\mathbf{w}_0$ are **parameters** adjusted by backpropagation and gradient descent

From deterministic to Bayesian

Bayesian view (Neal 1994)

Weights become **random variables** with a prior distribution:

Output layer weights:

$$w_{1j} \sim \mathcal{N}\left(0, \frac{\alpha}{J}\right), \quad w_{10} \sim \mathcal{N}(0, \sigma^2)$$

Hidden layer weights:

$$w_{ji} \sim \mathcal{N}(0, \sigma_w^2), \quad w_{j0} \sim \mathcal{N}(0, \sigma_b^2)$$

From deterministic to Bayesian

What Changes?

In the Bayesian view, the **network output becomes a random variable**:

$$f(\mathbf{x}) \sim ?$$

- The **input** is a fixed point $\mathbf{x}$
- **Randomness** is added by random weights
- The **output** is a random scalar $f(\mathbf{x})$

Central Limit Theorem

For any fixed finite set of inputs $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$:

The vector of outputs:

$$(f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(N)}))$$

can be written as a sum of J i.i.d. random vectors (one contribution per hidden unit).

By the multivariate Central Limit Theorem:

$$\lim_{J \rightarrow \infty} (f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(N)})) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

where $\boldsymbol{\mu} = \mathbf{0}$ and $\boldsymbol{\Sigma}_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$.

Infinite Neural Network = Gaussian Process

Theorem (Neal 1994)

When a one-hidden-layer Bayesian neural network has $J \rightarrow \infty$ hidden units with i.i.d. zero-mean weight priors (properly scaled), the random function converges in distribution to a Gaussian Process:

$$f(\cdot) \sim \mathcal{GP} \left(0, k(\mathbf{x}, \mathbf{x}') = \alpha \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) h \left(\sum_i w_{ji} x'_i + w_{j0} \right) \right] + \sigma^2 \right)$$

Mean Function

$$\begin{aligned} m(\mathbf{x}) &= \mathbb{E} \left[\sum_{j=1}^J w_{1j} h \left(\sum_i w_{ji} x_i + w_{j0} \right) + w_{10} \right] \\ &= \sum_{j=1}^J \mathbb{E}[w_{1j}] \cdot \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) \right] + \mathbb{E}[w_{10}] \\ &= \sum_{j=1}^J 0 \cdot \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) \right] + 0 \\ &= 0 \end{aligned}$$

where we used that all weights have **zero mean**.

Covariance Function

For any two distinct inputs $\mathbf{x}, \mathbf{x}'$:

$$k(\mathbf{x}, \mathbf{x}') = \text{Cov} [f(\mathbf{x}), f(\mathbf{x}')] = \mathbb{E} [f(\mathbf{x}) \cdot f(\mathbf{x}')]]$$

(since $\mathbb{E}[f(\mathbf{x})] = 0$)

Covariance Function Derivation

$$\begin{aligned}k(\mathbf{x}, \mathbf{x}') &= \mathbb{E} \left[\left(\sum_{j=1}^J w_{1j} h \left(\sum_i w_{ji} x_i + w_{j0} \right) + w_{10} \right) \left(\sum_{j=1}^J w_{1j} h \left(\sum_i w_{ji} x'_i + w_{j0} \right) + w_{10} \right) \right] \\&= \mathbb{E} \left[\sum_{j=1}^J w_{1j}^2 h \left(\sum_i w_{ji} x_i + w_{j0} \right) h \left(\sum_i w_{ji} x'_i + w_{j0} \right) \right] + \mathbb{E}[w_{10}^2] \\&= \sum_{j=1}^J \mathbb{E}[w_{1j}^2] \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) h \left(\sum_i w_{ji} x'_i + w_{j0} \right) \right] + \text{Var}(w_{10}) \\&= \sum_{j=1}^J \frac{\alpha}{J} \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) h \left(\sum_i w_{ji} x'_i + w_{j0} \right) \right] + \sigma^2 \\&= \alpha \mathbb{E} \left[h \left(\sum_i w_{ji} x_i + w_{j0} \right) h \left(\sum_i w_{ji} x'_i + w_{j0} \right) \right] + \sigma^2\end{aligned}$$

Extension to Deep Networks

Published as a conference paper at ICLR 2018

DEEP NEURAL NETWORKS AS GAUSSIAN PROCESSES

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Extension to Deep Networks

Single hidden layer to multiple layers

The Central Limit Theorem argument extends by **induction** over layers. If the input to layer ℓ is governed by a GP, then:

- The output of layer ℓ is a **sum of independent terms** (one per neuron).
- By CLT, it is also a GP.
- This applies recursively to all layers.

Result: An **infinitely wide deep network** (all layers with $N_\ell \rightarrow \infty$) is a Gaussian Process.

Recursive kernel computation

Building kernels layer by layer

For a deep network with L hidden layers, denote:

- $z^{(\ell)}(x)$: output of layer ℓ (pre-activation of next layer)
- $x^{(\ell)}(x)$: activations of layer ℓ (post-nonlinearity)

Layer ℓ output:

$$z_i^{(\ell)}(x) = b_i^{(\ell)} + \sum_{j=1}^{N_\ell} W_{ij}^{(\ell)} x_j^{(\ell-1)}(x)$$

where $x_j^{(\ell-1)}(x) = \varphi(z_j^{(\ell-1)}(x))$ (activation function φ).

Recursive kernel formula

Base case (input layer):

$$K^{(0)}(x, x') = \mathbb{E}[z^{(0)}(x)z^{(0)}(x')]$$

For raw inputs:

$$K^{(0)}(x, x') = \sigma_b^2 + \sigma_w^2 \frac{\mathbf{x} \cdot \mathbf{x}'}{d_{in}}$$

where σ_w^2, σ_b^2 are weight and bias variances, d_{in} is input dimension.

Recursive kernel: inductive step

Assume layer $\ell - 1$ computes a GP with kernel $K^{(\ell-1)}$.

For layer ℓ , the output is:

$$z_i^{(\ell)}(x) = b_i^{(\ell)} + \sum_{j=1}^{N_\ell} W_{ij}^{(\ell)} \varphi(z_j^{(\ell-1)}(x))$$

As $N_\ell \rightarrow \infty$, by CLT:

$$z^{(\ell)}(x) \sim \mathcal{GP}(0, K^{(\ell)})$$

with kernel:

$$K^{(\ell)}(x, x') = \sigma_b^2 + \sigma_w^2 \mathbb{E}_{z^{(\ell-1)} \sim \mathcal{GP}(0, K^{(\ell-1)})} [\varphi(z^{(\ell-1)}(x)) \varphi(z^{(\ell-1)}(x'))]$$

Gaussian Process Deconvolution

Gaussian Process Deconvolution

Application	Domain
De-reverberation	Acoustic signal processing
Super-resolution	Image restoration, low-resolution to high-resolution
Perfusion imaging	fMRI, medical imaging (hemodynamic response)
Seismic imaging	Geophysics, signal propagation
Astronomy	Point-spread function deconvolution

The Deconvolution Problem

We observe noisy, possibly incomplete measurements y of a convolved signal:

$$y(t) = \int h(\tau)x(t - \tau)d\tau + \epsilon(t)$$

where:

- $x(t)$ is the latent source signal.
- $h(\tau)$ is the convolution filter (known or unknown).
- $\epsilon(t)$ is the measurement noise.
- $y(t)$ is the observed signal

Challenges

Traditional deconvolution methods (Wiener, Inverse FT) often suffer from:

- **Numerical instability** when filter has low spectral power.
- **No uncertainty quantification** (only point estimates).
- **Assumptions on signal structure** that may be unrealistic.

The GP Solution

Place a **Gaussian process prior** on the latent source $x(t)$.

The GP Hierarchical Model:

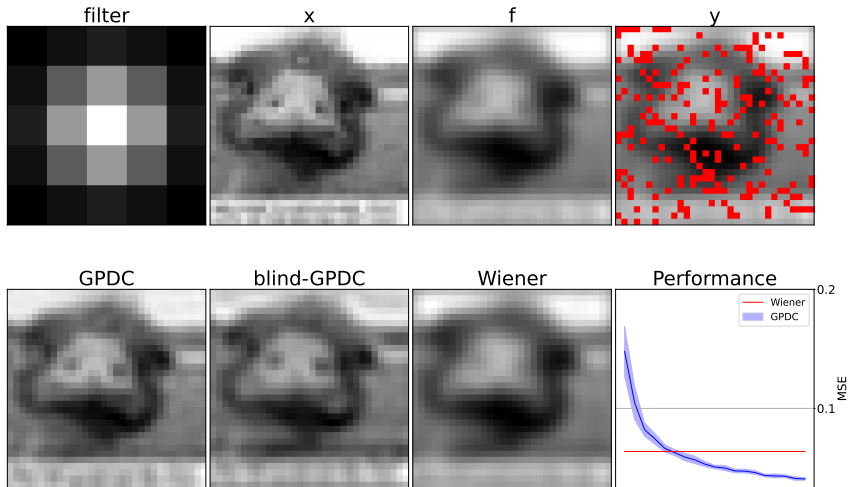
Given GP prior: $x(t) \sim \mathcal{GP}(m(t), k(t, t'))$

The posterior deconvolution is:

$$\hat{x}|y \sim \mathcal{GP}(\hat{m}(t), \hat{k}(t, t'))$$

where covariances are derived through convolution operators applied to the kernel k .

Example result



Resources

Rasmussen, Carl Edward. Gaussian processes in machine learning. Springer, Berlin, Heidelberg, 2003.

Richard Turner lecture notes

Mauricio Álvarez lecture notes

Gaussian processes in Python - scikit-learn.

Gaussian processes in Python - GPy.

Gaussian processes in Python - Gpytorch.

Gaussian process deconvolution (GAMES-UChile)